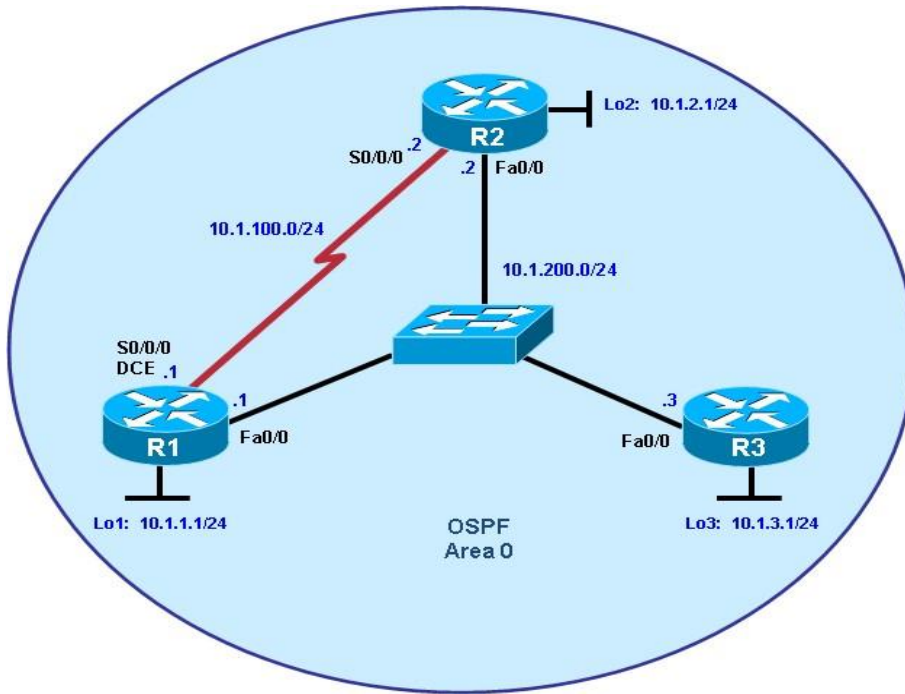


a.) Single-Area OSPF Link Costs and Interface Priorities

Topology



Objectives

- Configure single-area OSPF on a router.
- Advertise loopback interfaces into OSPF.
- Verify OSPF adjacencies.
- Verify OSPF routing information exchange.
- Modify OSPF link costs.
- Change interface priorities.
- Utilize debugging commands for troubleshooting OSPF.

Background

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- 1 switch (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- Serial and Ethernet cables

Step 1: Configure addressing and loopbacks.

- Using the addressing scheme in the diagram, apply IP addresses to the Fast Ethernet interfaces on R1, R2, and R3. Create Loopback1 on R1, Loopback2 on R2, and Loopback3 on R3, and address them according to the diagram.

```
R1# configure terminal
R1(config)# interface Loopback1
R1(config-if)# description Engineering Department R1(config-if)#
ip address 10.1.1.1 255.255.255.0
```

```

R1(config-if)# exit
R1(config)# interface FastEthernet0/0
R1(config-if)# ip address 10.1.200.1 255.255.255.0
R1(config-if)# no shutdown

```

- b. Configure the serial interfaces on R1 and R2 with the IP addresses shown in the diagram. Add the **clockrate** command where needed.

```

R1(config)# interface Serial 0/0/0
R1(config-if)# ip address 10.1.100.1 255.255.255.0
R1(config-if)# clockrate 64000 R1(config-
if)# bandwidth 64 R1(config-if)# no
shutdown

```

Step 2: Add physical interfaces to OSPF.

- a. Enter the commands on R1. Exit to privileged EXEC mode and type **debug ip ospf adj**. The **debug** command lets you watch OSPF neighbors come up and see neighbor relationships.

```

R1(config)# router ospf 1
R1(config-router)# network 10.1.100.0 0.0.0.255 area 0
R1(config-router)# network 10.1.200.0 0.0.0.255 area 0
R1(config-router)# end
R1#

```

```

R1# debug ip ospf adj
OSPF adjacency events debugging is on

```

=d the level of control given by wildcard masks to assign different interfaces to their appropriate areas and not restrict an entire major network to be in one area. There might be networks connected to a router that the administrator does not want to advertise but which are in the same major network as the OSPF-enabled interface. Without using wildcard masks, it would be practically impossible to implement this.

Step 3: Use OSPF show commands.

- a. The **show ip protocols** command displays basic high-level routing protocol information. The output lists each OSPF process, the router ID, and which networks OSPF is routing for in each area. This information can be useful in debugging routing operations.

```

R1# show ip protocols
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.1.100.0 0.0.0.255 area 0
    10.1.200.1 0.0.0.0 area 0
  Reference bandwidth unit is 100 mbps
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)

```

- b. The **show ip ospf** command displays the OSPF process ID and router ID.

```

R1# show ip ospf
Routing Process "ospf 1" with ID 10.1.1.1
  Start time: 00:17:44.612, Time elapsed: 00:10:51.408
  Supports only single TOS(TOS0) routes
  Supports opaque LSA

```

```

Supports Link-local Signaling (LLS)
Supports area transit capability
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msec
Minimum hold time between two consecutive SPF's 10000 msec
Maximum wait time between two consecutive SPF's 10000 msec
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 2
    Area has no authentication
    SPF algorithm last executed 00:03:21.132 ago
    SPF algorithm executed 5 times
    Area ranges are
    Number of LSA 4. Checksum Sum 0x021A30
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

```

- c. The **show ip ospf neighbor** command displays important neighbor status, including the adjacency state, address, router ID, and connected interface.

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.1.2.1	1	FULL/BDR	00:00:36	10.1.200.2	FastEthernet0/0
10.1.3.1	1	FULL/DR	00:00:35	10.1.200.3	FastEthernet0/0
10.1.2.1	0	FULL/ -	00:00:36	10.1.100.2	Serial0/0/0

If you need more detail than the standard one-line summaries of neighbors, use the **show ip ospf neighbor detail** command. However, generally, the regular command gives you all that you need.

- d. The **show ip ospf interface interface_type number** command shows interface timers and network types.

```
R1# show ip ospf interface FastEthernet 0/0
```

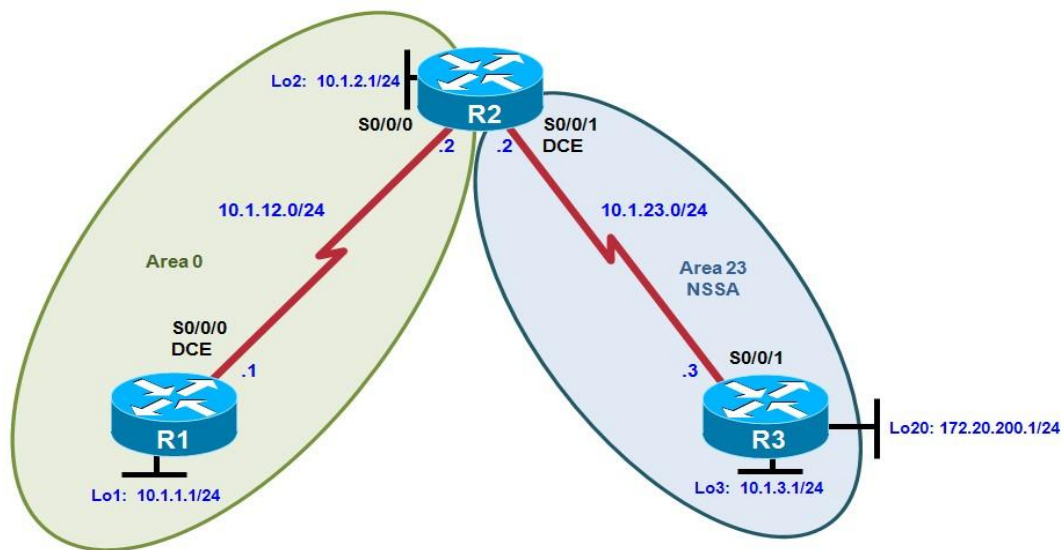
```

FastEthernet0/0 is up, line protocol is up
  Internet Address 10.1.200.1/24, Area 0
  Process ID 1, Router ID 10.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DROTHER, Priority 1
  Designated Router (ID) 10.1.3.1, Interface address 10.1.200.3
  Backup Designated router (ID) 10.1.2.1, Interface address 10.1.200.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5

```

b) Multi-Area OSPF with Stub Areas and Authentication

Topology



Objectives

- Configure multiple-area OSPF on a router.
- Verify multiple-area behavior.
- Configure OSPF stub, totally stubby, and not-so-stubby areas.
- Configure OSPF authentication.

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- Serial and console cables

Step 1: Configure addressing and loopbacks.

- Using the addressing scheme in the diagram, apply IP addresses to the serial interfaces on R1, R2, and R3. Create loopbacks on R1, R2, and R3, and address them according to the diagram.

Note: Depending on the router models you have, you might need to add clock rates to the DCE end of each connection (newer equipment adds this automatically). Verify connectivity across each serial link.

```
R1# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# interface loopback 1
R1(config-if)# description Engineering Department
R1(config-if)# ip address 10.1.1.1 255.255.255.0
R1(config-if)# interface serial 0/0/0
R1(config-if)# ip address 10.1.12.1 255.255.255.0
R1(config-if)# clockrate 64000
R1(config-if)# no shutdown
```

Step 2: Add interfaces into OSPF.

- Create OSPF process 1 on routers R1 and R2. Configure the subnet of the serial link between R1 and R2 to be in OSPF area 0 using the **network** command. Add loopback 1 on R1 and loopback 2 on R2 into OSPF

area 0. Change the network type on the loopback interfaces so that they are advertised with the correct subnet.

```
R1(config)# router ospf 1
R1(config-router)# network 10.1.12.0 0.0.0.255 area 0
R1(config-router)# network 10.1.1.0 0.0.0.255 area 0
R1(config-router)# exit
R1(config)# interface loopback 1
R1(config-if)# ip ospf network point-to-point
```

- b. Verify that both routers have OSPF neighbors using the **show ip ospf neighbors** command. R1# **show ip ospf neighbor**

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.1.2.1	0	FULL/ -	00:00:38	10.1.12.2	Serial0/0/0

R2# **show ip ospf neighbor**

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.1.1.1	0	FULL/ -	00:00:35	10.1.12.1	Serial0/0/0

- c. Verify that the routers can see each other's loopback with the **show ip route** command.

R1# **show ip route**

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/24 is subnetted, 3 subnets
C      10.1.12.0 is directly connected, Serial0/0/0
O      10.1.2.0 [110/65] via 10.1.12.2, 00:00:10, Serial0/0/0
C      10.1.1.0 is directly connected, Loopback1
```

R2# **show ip route**

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/24 is subnetted, 4 subnets
C      10.1.12.0 is directly connected, Serial0/0/0
C      10.1.2.0 is directly connected, Loopback2
O      10.1.1.0 [110/65] via 10.1.12.1, 00:00:30, Serial0/0/0 C
10.1.23.0 is directly connected, Serial0/0/1
```

- d. Add the subnet between R2 and R3 into OSPF area 23 using the **network** command. Add loopback 3 on R3 into area 23.

```
R2(config)# router ospf 1
R2(config-router)# network 10.1.23.0 0.0.0.255 area 23
```

- e. Verify that this neighbor relationship comes up using the **show ip ospf neighbors** command. R2# **show ip ospf neighbor**

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.1.1.1	0	FULL/ -	00:00:36	10.1.12.1	Serial0/0/0
10.1.3.1	0	FULL/ -	00:00:36	10.1.23.3	Serial0/0/1

- f. If you look at the output of the **show ip route** command on R1, you see a route to the R3 loopback. Notice that it is identified as an inter-area route.

```
R1# show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/24 is subnetted, 5 subnets
C      10.1.12.0 is directly connected, Serial0/0/0
O IA   10.1.3.0 [110/129] via 10.1.12.2, 00:00:28, Serial0/0/0
O      10.1.2.0 [110/65] via 10.1.12.2, 00:01:38, Serial0/0/0 C
10.1.1.0 is directly connected, Loopback1
O IA   10.1.23.0 [110/128] via 10.1.12.2, 00:01:38, Serial0/0/0

       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Step 3: Configure a stub area.

- a. Under the OSPF process on R2 and R3, make area 23 the stub area using the **area area stub** command. The adjacency between the two routers might go down during the transition period, but it should come back up afterwards.

```
R2(config)# router ospf 1
R2(config-router)# area 23 stub
```

- b. Confirm that it comes up by using the **show ip ospf neighbors** command.

```
R2# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.1.1.1	0	FULL/ -	00:00:36	10.1.12.1	Serial0/0/0
10.1.3.1	0	FULL/ -	00:00:36	10.1.23.3	Serial0/0/1

Step 4: Configure a totally stubby area.

- a. To see how this works, issue the **show ip route** command on R3. Notice the inter-area routes, in addition to the default route generated by R2.

```
R3# show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
```

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.1.23.2 to network 0.0.0.0

172.20.0.0/24 is subnetted, 1 subnets
C 172.20.200.0 is directly connected, Loopback20
10.0.0.0/24 is subnetted, 5 subnets
O IA 10.1.12.0 [110/128] via 10.1.23.2, 00:00:56, Serial0/0/1 C
10.1.3.0 is directly connected, Loopback3
O IA 10.1.2.0 [110/65] via 10.1.23.2, 00:00:56, Serial0/0/1
O IA 10.1.1.0 [110/129] via 10.1.23.2, 00:00:56, Serial0/0/1 C
10.1.23.0 is directly connected, Serial0/0/1
O*IA 0.0.0.0/0 [110/65] via 10.1.23.2, 00:00:56, Serial0/0/1

- b. Look at the output of the **show ip ospf database** command on R2 to see which LSAs are in its OSPF database.

R2# **show ip ospf database**

OSPF Router with ID (10.1.2.1) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.1.1.1	10.1.1.1	435	0x80000004	0x0056D6	3
10.1.2.1	10.1.2.1	358	0x80000003	0x0057D2	3

Summary Net Link States (Area 0)

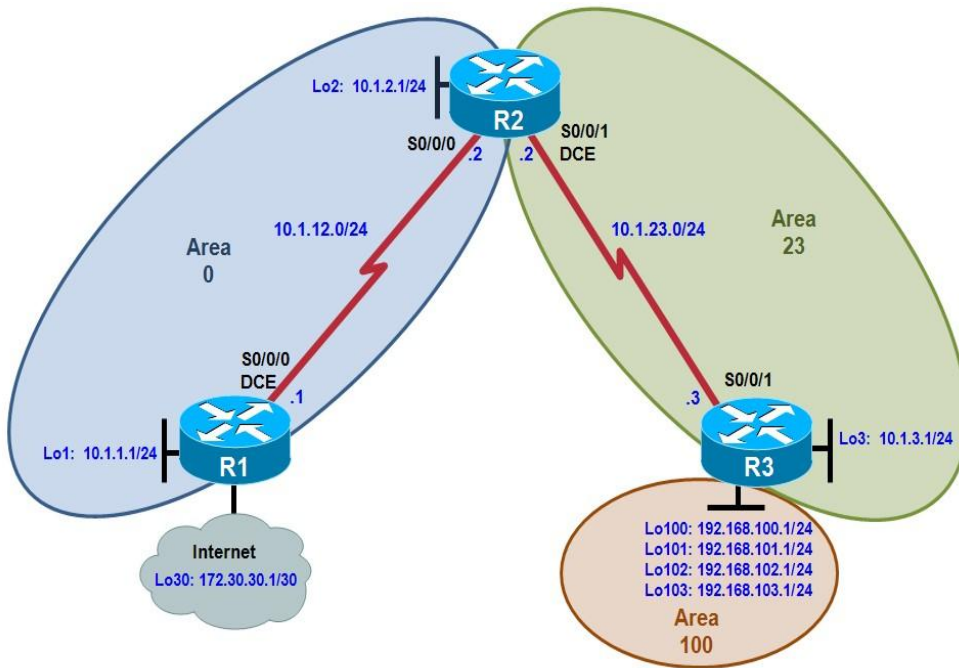
Link ID	ADV Router	Age	Seq#	Checksum
10.1.3.0	10.1.2.1	174	0x80000001	0x00EFEF
10.1.23.0	10.1.2.1	354	0x80000001	0x0009C3

Router Link States (Area 23)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
---------	------------	-----	------	----------	------------

Practical 2

a), OSPF Virtual Links and Area Summarization Topology



Objectives

- Configure multi-area OSPF on a router.
- Verify multi-area behavior.
- Create an OSPF virtual link.
- Summarize an area.
- Generate a default route into OSPF.

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- Serial and console cables

Step 1: Configure addressing and loopbacks.

Using the addressing scheme in the diagram, apply IP addresses to the serial interfaces on R1, R2, and R3. Create loopbacks on R1, R2, and R3, and address them according to the diagram.

```
R1# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# interface loopback 1
R1(config-if)# description Engineering Department R1(config-if)# ip address 10.1.1.1 255.255.255.0
R1(config-if)# interface loopback 30
R1(config-if)# ip address 172.30.30.1 255.255.255.252
R1(config-if)# interface serial 0/0/0
R1(config-if)# ip address 10.1.12.1 255.255.255.0
R1(config-if)# clockrate 64000 R1(config-if)# no shutdown
```


Step 2: Add interfaces into OSPF.

- a. Create OSPF process 1 on all three routers. Using the **network** command, configure the subnet of the serial link between R1 and R2 to be in OSPF area 0

```
R1(config)# router ospf 1
R1(config-router)# network 10.1.12.0 0.0.0.255 area 0
R1(config-router)# network 10.1.1.0 0.0.0.255 area 0
R1(config-router)# exit
R1(config)# interface loopback 1
R1(config-if)# ip ospf network point-to-point
```

- b. Verify that you can see OSPF neighbors in the **show ip ospf neighbors** output on both routers. Verify that the routers can see each other's loopback with the **show ip route** command.

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.1.2.1	0	FULL/ -	00:00:38	10.1.12.2	Serial0/0/0

```
R1# show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/24 is subnetted, 3 subnets
C      10.1.12.0 is directly connected, Serial0/0/0
O      10.1.2.0 [110/65] via 10.1.12.2, 00:00:10, Serial0/0/0
C      10.1.1.0 is directly connected, Loopback1
172.30.0.0/30 is subnetted, 1 subnets
C      172.30.30.0 is directly connected, Loopback30
```

- c. Add the subnet between R2 and R3 into OSPF area 23 using the **network** command. Add loopback 3 on R3 into area 23.

```
R2(config)# router ospf 1
R2(config-router)# network 10.1.23.0 0.0.0.255 area 23
```

```
R3(config)# router ospf 1
R3(config-router)# network 10.1.23.0 0.0.0.255 area 23
R3(config-router)# network 10.1.3.0 0.0.0.255 area 23
R3(config-router)# exit
```

Step 3: Create a virtual link.

- a. Add loopbacks 100 through 103 on R3 to the OSPF process in area 100 using the **network** command. Change the network type to advertise the correct subnet mask.

```
R3(config)# router ospf 1
R3(config-router)# network 192.168.100.0 0.0.3.255 area 100
R3(config-router)# exit
R3(config)# interface loopback 100
R3(config-if)# ip ospf network point-to-point
```

```

R3(config-if)# interface loopback 101
R3(config-if)# ip ospf network point-to-point
R3(config-if)# interface loopback 102
R3(config-if)# ip ospf network point-to-point
R3(config-if)# interface loopback 103
R3(config-if)# ip ospf network point-to-point

```

- c. Create a virtual link using the `area transit_area virtual-link router-id` OSPF configuration command on both R2 and R3.

```

R2(config)# router ospf 1
R2(config-router)# area 23 virtual-link 192.168.103.1

```

- d. After you see the adjacency over the virtual interface come up, issue the `show ip route` command on R2 and see the routes from area 100. You can verify the virtual link with the `show ip ospf neighbor` and `show ip ospf interface` commands.

```

R2# show ip route

```

```

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route
Gateway of last resort is not set
 10.0.0.0/24 is subnetted, 5 subnets
C       10.1.12.0 is directly connected, Serial0/0/0
O       10.1.3.0 [110/65] via 10.1.23.3, 00:01:35, Serial0/0/1
C       10.1.2.0 is directly connected, Loopback2
O       10.1.1.0 [110/65] via 10.1.12.1, 00:01:35, Serial0/0/0
C       10.1.23.0 is directly connected, Serial0/0/1
O IA 192.168.102.0/24 [110/65] via 10.1.23.3, 00:00:05, Serial0/0/1
O IA 192.168.103.0/24 [110/65] via 10.1.23.3, 00:00:05, Serial0/0/1
O IA 192.168.100.0/24 [110/65] via 10.1.23.3, 00:00:57, Serial0/0/1
O IA 192.168.101.0/24 [110/65] via 10.1.23.3, 00:00:16, Serial0/0/1

```

```

R2# show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.103.1	0	FULL/ -	-	10.1.23.3	OSPF_VL0
172.30.30.1	0	FULL/ -	00:00:30	10.1.12.1	Serial0/0/0
192.168.103.1	0	FULL/ -	00:00:30	10.1.23.3	Serial0/0/1

```

R2# show ip ospf interface

```

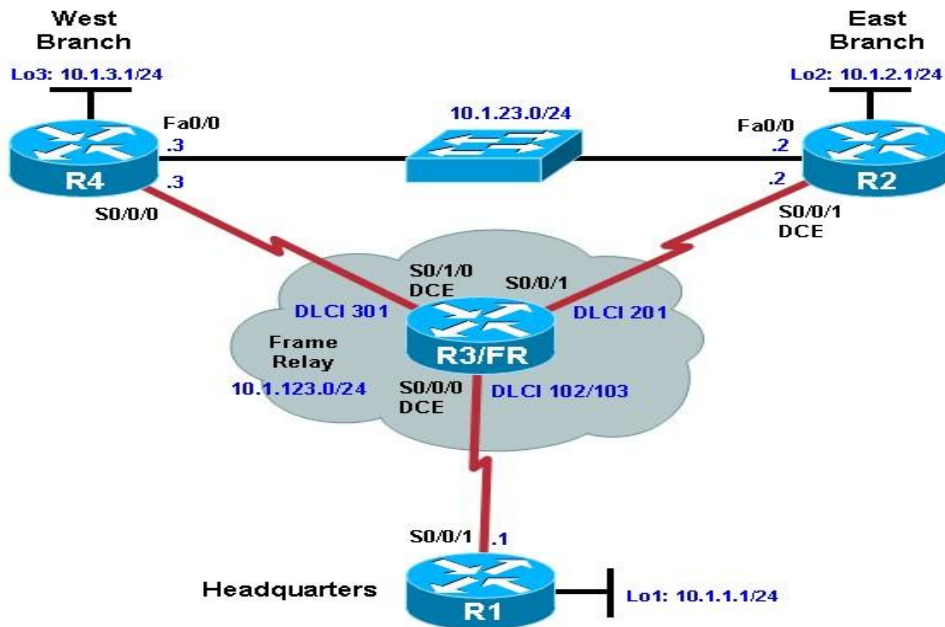
```

OSPF_VL0 is up, line protocol is up
Internet Address 10.1.23.2/24, Area 0
Process ID 1, Router ID 10.1.2.1, Network Type VIRTUAL_LINK, Cost: 64
Configured as demand circuit.
Run as demand circuit.
DoNotAge LSA allowed.
Transmit Delay is 1 sec, State POINT_TO_POINT,
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  oob-resync timeout 40
Hello due in 00:00:03
Supports Link-local Signaling (LLS)

```

b) OSPF over Frame Relay

Topology



Objectives

- Configure OSPF over Frame Relay.
- Use non-broadcast and point-to-multipoint OSPF network types.
- Modify default OSPF timers.

Required Resources

- 4 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- 1 switch (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- Serial and Ethernet cables

Step 1: Configure Frame Relay and addressing on the HQ, EAST, and WEST routers.

- Configure the router physical interfaces with IP addresses.

. Router R1 (Hostname HQ)

Router# **configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

```
Router(config)# hostname HQ
HQ(config)# interface loopback 1
HQ(config-if)# ip address 10.1.1.1 255.255.255.0
HQ(config-if)# interface serial 0/0/1
HQ(config-if)# ip address 10.1.123.1 255.255.255.0
HQ(config-if)# encapsulation frame-relay ietf
HQ(config-if)# no frame-relay inverse-arp
HQ(config-if)# frame-relay map ip 10.1.123.1 102
HQ(config-if)# frame-relay map ip 10.1.123.2 102
```

```
HQ(config-if)# frame-relay map ip 10.1.123.3 103  
HQ(config-if)# no shutdown
```

Step 2: Configure the Frame Relay switch.

Configure the Frame Relay switch to have the DLCIs indicated in the diagram between HQ and EAST and HQ and WEST.

```
Router R3 (FRS) hostname  
FRS ! frame-relay  
switching interface  
Serial0/0/0 no ip  
address  
encapsulation frame-relay ietf  
no ip route-cache  
clock rate 64000 frame-  
relay intf-type dce  
frame-relay route 102 interface Serial0/0/1 201  
frame-relay route 103 interface Serial0/1/0 301 no  
shutdown  
! interface  
Serial0/0/1 no ip  
address  
encapsulation frame-relay ietf frame-  
relay intf-type dce  
frame-relay route 201 interface Serial0/0/0 102  
no shutdown  
! interface  
Serial0/1/0 no ip  
address  
encapsulation frame-relay ietf  
no ip route-cache frame-relay  
intf-type dce clock rate 64000  
frame-relay route 301 interface Serial0/0/0 103  
no shutdown
```

Step 3: Configure OSPF network type NBMA.

a. Create OSPF process 1. Add the Frame Relay interfaces on each router into area 0 with the **network** command. Add the loopback interfaces on each router into area 0, and then change the network type to allow the correct subnet mask to be advertised.

```
HQ(config)# router ospf 1  
HQ(config-router)# network 10.1.123.0 0.0.0.255 area 0  
HQ(config-router)# network 10.1.1.0 0.0.0.255 area 0  
HQ(config-router)# exit  
HQ(config)# interface loopback 1  
HQ(config-if)# ip ospf network point-to-point
```

b. You can verify adjacency states with the **show ip ospf neighbor** command.

```
HQ# show ip ospf neighbor
```

Step 4: Change the network type to point-to-multipoint.

a. To configure point-to-multipoint links, remove the previously configured **neighbor** statements and interface priorities..

```
HQ(config)# router ospf 1  
HQ(config-router)# no neighbor 10.1.123.2
```

```
HQ(config-router)# no neighbor 10.1.123.3
```

- b. Use the interface command **ip ospf network point-to-multipoint** and reapply the Frame Relay maps using the **broadcast** option.

```
EAST(config)# interface serial 0/0/1
EAST(config-if)# ip ospf network point-to-multipoint
EAST(config-if)# frame-relay map ip 10.1.123.1 201 broadcast
EAST(config-if)# frame-relay map ip 10.1.123.3 201 broadcast
```

```
HQ# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.1.3.1	0	FULL/ -	00:01:34	10.1.123.3	Serial0/0/1
10.1.2.1	0	FULL/ -	00:01:45	10.1.123.2	Serial0/0/1

- c. Observe the routing table on one of the spoke routers. Notice how the routing table has host routes in it. This is part of point-to-multipoint behavior.

```
EAST# show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2      E1
- OSPF external type 1, E2 - OSPF external type 2
  i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
  ia - IS-IS inter area, * - candidate default, U - per-user static
route
  o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O      10.1.3.0/24 [110/129] via 10.1.123.1, 00:01:07, Serial0/0/1 C
10.1.2.0/24 is directly connected, Loopback2
O      10.1.1.0/24 [110/65] via 10.1.123.1, 00:01:07, Serial0/0/1
C      10.1.23.0/24 is directly connected, FastEthernet0/0
C      10.1.123.0/24 is directly connected, Serial0/0/1
O      10.1.123.1/32 [110/64] via 10.1.123.1, 00:01:07, Serial0/0/1
O      10.1.123.3/32 [110/128] via 10.1.123.1, 00:01:07, Serial0/0/1
```

- d. Look at the output of the **show ip ospf interface interface** command on your routers. Notice that the interface type is point-to-multipoint.

```
EAST# show ip ospf interface serial 0/0/1
```

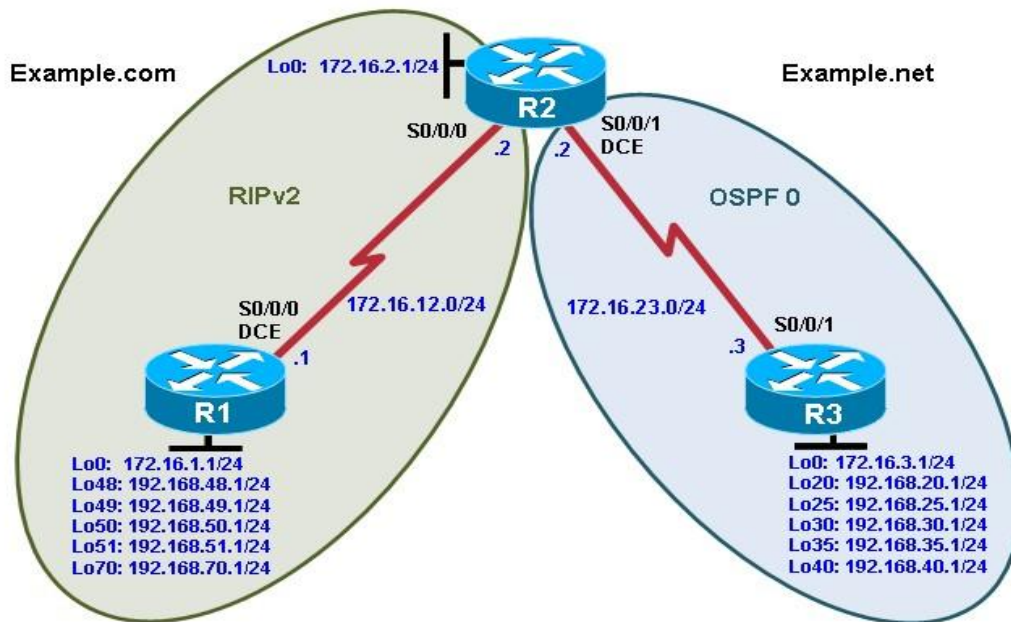
Step 5: Change OSPF timers.

- a. Add the Ethernet link connecting EAST and WEST to the OSPF process using the **network** command.

```
EAST(config)# router ospf 1
EAST(config-router)# network 10.1.23.0 0.0.0.255 area 0
```

Practical 3

a) Redistribution Between RIP and OSPF Topology



Objectives

- Review configuration and verification of RIP and OSPF.
- Configure passive interfaces in both RIP and OSPF.
- Filter routing updates using distribute lists.
- Redistribute static routes into RIP.
- Redistribute RIP routes into OSPF.
- Redistribute OSPF routes into RIP.
- Originate a default route into OSPF.
- Set a default seed metric.
- Modify OSPF external network types.
- Configure summary addresses.

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- Serial and console cables

Step 1: Configure loopbacks and assign addresses.

- Configure all loopback interfaces on the three routers in the diagram. Configure the serial interfaces with the IP addresses, bring them up, and set a DCE clock rate where appropriate.

```
R1(config)# interface Loopback0
R1(config-if)# ip address 172.16.1.1 255.255.255.0
R1(config-if)# interface Loopback48
R1(config-if)# ip address 192.168.48.1 255.255.255.0
R1(config-if)# interface Loopback49
R1(config-if)# ip address 192.168.49.1 255.255.255.0
R1(config-if)# interface Loopback50
```

```

R1(config-if)# ip address 192.168.50.1 255.255.255.0
R1(config-if)# interface Loopback51
R1(config-if)# ip address 192.168.51.1 255.255.255.0
R1(config-if)# interface Loopback70
R1(config-if)# ip address 192.168.70.1 255.255.255.0
R1(config-if)# interface Serial0/0/0
R1(config-if)# ip address 172.16.12.1 255.255.255.0
R1(config-if)# clock rate 64000 R1(config-
if)# bandwidth 64 R1(config-if)# no
shutdown

```

- b. (Optional) On each router, create an enable secret password. Configure the console line for synchronous logging and no timeout. Configure the vty lines to allow Telnet to and remote configuration of network devices. R1 example:

```

R1(config)# enable secret cisco

```

```

R1(config)# line con 0
R1(config-line)# logging synchronous
R1(config-line)# exec-timeout 0 0

```

```

R1(config)# line vty 0 4
R1(config-line)# password cisco R1(config-line)#
login

```

Step 2: Configure RIPv2.

Configuring RIPv2 on a router is fairly simple:

- Type the global configuration command **router rip** to enter RIP configuration mode.
 - Enable RIPv2 with the **version 2** command.
 - Enter the **no auto-summary** command to disable automatic summarization at classful network boundaries.
 - Add the networks you want using the **network network** command.
- b. Verify that the RIP routes were learned from the other routers using the **show ip route rip** command on each router.

```

R1# show ip route rip
      172.16.0.0/24 is subnetted, 4 subnets
R       172.16.23.0 [120/1] via 172.16.12.2, 00:00:03, Serial0/0/0
R       172.16.2.0 [120/1] via 172.16.12.2, 00:00:03, Serial0/0/0

```

- c. You can also verify which routes are coming in from RIP advertisements with the **show ip rip database** command.

```

R1# show ip rip database
172.16.0.0/16      auto-summary
172.16.1.0/24      directly connected, Loopback0
172.16.2.0/24      [1] via 172.16.12.2, 00:00:06,
Serial0/0/0
172.16.12.0/24     directly
connected, Serial0/0/0
172.16.23.0/24     [1] via
172.16.12.2,

```

```

00:00:06, Serial0/0/0 192.168.48.0/24    auto-summary
192.168.48.0/24    directly connected, Loopback48
192.168.49.0/24    auto-summary
192.168.49.0/24    directly connected, Loopback49
192.168.50.0/24    auto-summary
192.168.50.0/24    directly connected, Loopback50
192.168.51.0/24    auto-summary
192.168.51.0/24    directly connected, Loopback51
192.168.70.0/24    auto-summary
192.168.70.0/24    directly connected, Loopback70

```

Step 3: Configure passive interfaces in RIP.

- a. On R1, use the **show ip route rip** command to view the RIP routes in the routing table. Notice that the network for the serial interface of R2 that connects to R3 is present, even though you do not have a RIP neighbor on that interface. This is because the entire class B network 172.16.0.0 /16 was added to RIP on R2.

```
R1# show ip route rip
```

Step 4: Summarize a supernet with RIP.

- a. On R2, issue the **show ip route rip** command. Notice that you can see all prefixes from R1 in the R2 routing table.

```

R2# show ip route rip
    172.16.0.0/24 is subnetted, 4 subnets
R       172.16.1.0 [120/1] via 172.16.12.1, 00:00:29, Serial0/0/0
R       192.168.51.0/24 [120/1] via 172.16.12.1, 00:00:29, Serial0/0/0
R       192.168.50.0/24 [120/1] via 172.16.12.1, 00:00:29, Serial0/0/0
R       192.168.49.0/24 [120/1] via 172.16.12.1, 00:00:29, Serial0/0/0
R       192.168.70.0/24 [120/1] via 172.16.12.1, 00:00:29, Serial0/0/0
R       192.168.48.0/24 [120/1] via 172.16.12.1, 00:00:29, Serial0/0/0

```

```
R1(config)# interface serial 0/0/0
```

```
R1(config-if)# ip summary-address rip 192.168.48.0 255.255.252.0
```

Summary mask must be greater or equal to major net

- b. To get around the **ip summary-address rip** message error, create a static route on R1 to summarize the networks of loopbacks 48 through 51. Then redistribute the route on R1.

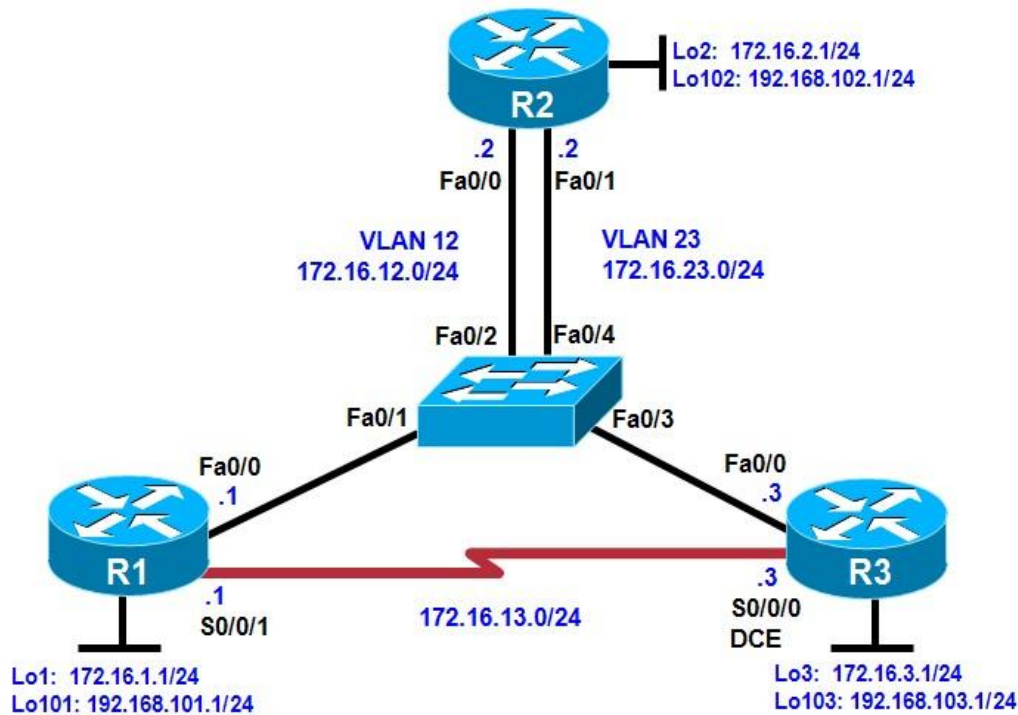
```
R1(config)# ip route 192.168.48.0 255.255.252.0 null0
```

```
R1(config)# router rip
```

```
R1(config-router)# redistribute static .
```


b). Manipulating Administrative Distances

Topology



Objectives

- Configure RIP on a router.
- Configure OSPF on a router.
- Manipulate administrative distances.
- Compare routing protocol behavior.

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- 1 switch (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)

Currently, EIGRP is considered the most trusted IGP on Cisco routers with an administrative distance of 90.

Step 2: Configure router loopbacks and addressing.

Configure all loopback interfaces on the three routers in the diagram. Configure the serial interface with the IP addresses, bring them up, and set a clock rate where appropriate.

```
R1# conf t
R1(config)# interface loopback 1
R1(config-if)# ip address 172.16.1.1 255.255.255.0
R1(config-if)# interface loopback 101
R1(config-if)# ip address 192.168.101.1 255.255.255.0
R1(config-if)# interface fastethernet 0/0
R1(config-if)# ip address 172.16.12.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# interface serial 0/0/1
R1(config-if)# bandwidth 64
```

```
R1(config-if)# ip address 172.16.13.1 255.255.255.0 R1(config-  
if)# no shutdown
```

Step 3: Configure switch VLANs.

- a. Configure the switch VLANs, and place the correct access ports in each VLAN.

```
Switch(config)# vlan 12  
Switch(config-vlan)# name R1-R2  
Switch(config-vlan)# vlan 23  
Switch(config-vlan)# name R2-R3  
Switch(config-vlan)# exit
```

Step 4: Configure RIP.

- a. Configure RIPv2 on all three routers for the major networks. Disable automatic summarization.

```
R1(config)# router rip  
R1(config-router)# version 2  
R1(config-router)# no auto-summary  
R1(config-router)# network 172.16.0.0  
R1(config-router)# network 192.168.101.0
```

- b. Verify the configuration using the **show ip route rip** command on each router.

```
R1# show ip route rip  
172.16.0.0/24 is subnetted, 6 subnets  
R    172.16.23.0 [120/1] via 172.16.13.3, 00:02:29, Serial0/0/1  
      [120/1] via 172.16.12.2, 00:02:15, FastEthernet0/0  
R    172.16.2.0 [120/1] via 172.16.12.2, 00:02:15, FastEthernet0/0  
R    172.16.3.0 [120/1] via 172.16.13.3, 00:02:29, Serial0/0/1  
R    192.168.102.0/24 [120/1] via 172.16.12.2, 00:02:15, FastEthernet0/0  
R    192.168.103.0/24 [120/1] via 172.16.13.3, 00:02:29, Serial0/0/1
```

- c. Verify that each router is receiving RIP routes from other routers using the **show ip protocols** command.

```
R1# show ip protocols  
Routing Protocol is "rip"  
Outgoing update filter list for all interfaces is not set  
Incoming update filter list for all interfaces is not set  
Sending updates every 30 seconds, next due in 26 seconds  
Invalid after 180 seconds, hold down 180, flushed after 240  
Redistributing: rip  
Default version control: send version 2, receive version 2  
Interface          Send  Recv  Triggered RIP  Key-chain  
FastEthernet0/0     2      2  
Serial0/0/1         2      2  
Loopback1           2      2  
Loopback101        2      2  
Automatic network summarization is not in effect  
Maximum path: 4  
Routing for Networks:  
172.16.0.0  
192.168.101.0  
Routing Information Sources:  
Gateway            Distance    Last Update  
172.16.12.2         120         00:00:21  
172.16.13.3         120         00:00:03  
Distance: (default is 120)
```

Step 5: Configure OSPF.

- a. Configure OSPF on all routers. Include the entire major network in area 0 on all three routers. Remember to change the network type on the loopback interfaces.

```
R1(config)# interface loopback 1
R1(config-if)# ip ospf network point-to-point
R1(config-if)# interface loopback 101
R1(config-if)# ip ospf network point-to-point
R1(config-if)# router ospf 1
R1(config-router)# network 172.16.0.0 0.0.255.255 area 0
R1(config-router)# network 192.168.101.0 0.0.0.255 area 0
```

- b. Verify the configuration using the **show ip ospf neighbors** and **show ip route** commands on each router.

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.103.1	0	FULL/ -	00:00:39	172.16.13.3	Serial0/0/1
192.168.102.1	1	FULL/DR	00:00:39	172.16.12.2	FastEthernet0/0

```
R1# show ip route
```

<output omitted>

```
172.16.0.0/24 is subnetted, 6 subnets
O       172.16.23.0 [110/2] via 172.16.12.2, 00:00:48, FastEthernet0/0
C       172.16.12.0 is directly connected, FastEthernet0/0
C       172.16.13.0 is directly connected, Serial0/0/1
C       172.16.1.0 is directly connected, Loopback1
O       172.16.2.0 [110/2] via 172.16.12.2, 00:00:48, FastEthernet0/0
O       172.16.3.0 [110/3] via 172.16.12.2, 00:00:48, FastEthernet0/0
O       192.168.102.0/24 [110/2] via 172.16.12.2, 00:00:48, FastEthernet0/0
O       192.168.103.0/24 [110/3] via 172.16.12.2, 00:00:49, FastEthernet0/0
C       192.168.101.0/24 is directly connected, Loopback101
```

Step 6: Modify the routing protocol distance.

The **distance** command is a protocol-independent way to manipulate routing protocol distances.

- a. On all three routers, change the distance of RIP to 100.

```
R1(config)# router rip
```

- b. Examine the output of the **show ip route** command. Notice that *all* the routes have become RIP routes because RIP now has a lower distance than OSPF.

```
R1# show ip route
```

Step 7: Modify distance based on route source.

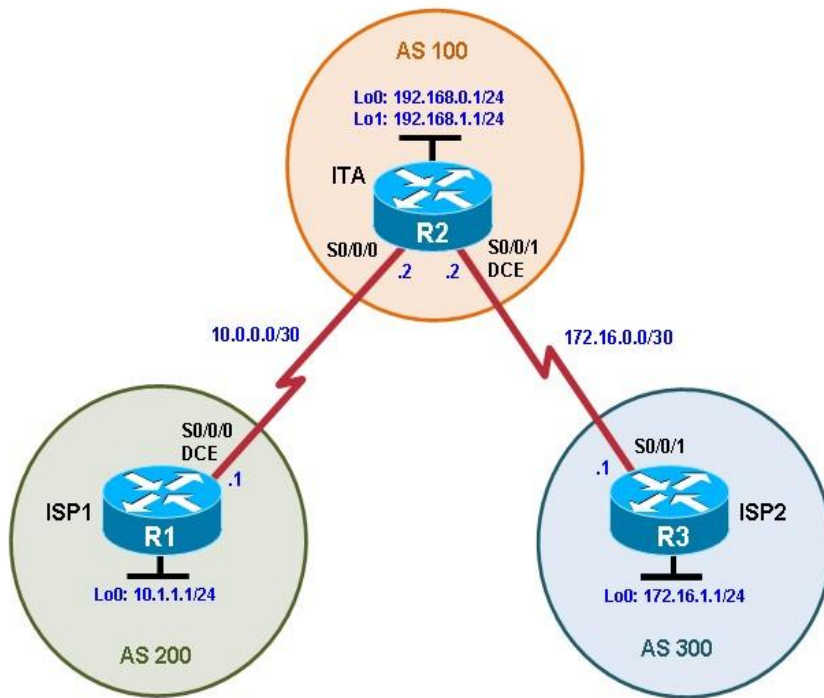
You can also modify administrative distance based on route source using the **distance distance**

- a. On all three routers, change the OSPF administrative distance to 85 for any routes being advertised from routers with IDs in the range of 192.168.100.0/21.

```
R1(config)# router ospf 1
```

Practical 4

a.) Configuring BGP with Default Routing Topology



Objectives

- Configure BGP to exchange routing information with two ISPs.

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- Serial and console cables

Step 1: Prepare the routers for the lab.

Cable the network as shown in the topology diagram. Erase the startup configuration and reload each router to clear previous configurations.

Step 2: Configure the hostname and interface addresses.

You can copy and paste the following configurations into your routers to begin.

```
Router R1 (hostname ISP1) hostname ISP1 ! interface Lo0
description ISP1 Internet Network
ip address 10.1.1.1 255.255.255.0 !
interface Serial0/0/0 description
ISP1 -> ITA ip address 10.0.0.1
255.255.255.252 clock rate 128000
no shutdown ! end
```

- b. Use **ping** to test the connectivity between the directly connected routers. Note that router ISP1 cannot reach router ISP2.

Step 3: Configure BGP on the ISP routers.

On the ISP1 and ISP2 routers, configure BGP to peer with the ITA boundary router and advertise the ISP loopback networks.

```
ISP1(config)# router bgp 200
ISP1(config-router)# neighbor 10.0.0.2 remote-as 100
ISP1(config-router)# network 10.1.1.0 mask 255.255.255.0

ISP2(config)# router bgp 300
ISP2(config-router)# neighbor 172.16.0.2 remote-as 100
ISP2(config-router)# network 172.16.1.0 mask 255.255.255.0
```

Step 4: Configure BGP on the ITA boundary router.

- a. Configure the ITA router to run BGP with both Internet providers.

```
ITA(config)# router bgp 100
ITA(config-router)# neighbor 10.0.0.1 remote-as 200
ITA(config-router)# neighbor 172.16.0.1 remote-as 300
ITA(config-router)# network 192.168.0.0
ITA(config-router)# network 192.168.1.0
```

- b. To verify the configuration, check the ITA routing table with the **show ip route** command.

```
ITA# show ip route
<output omitted>
```

```
Gateway of last resort is not set
172.16.0.0/16 is variably
subnetted, 2 subnets, 2 masks
C       172.16.0.0/30 is directly connected, Serial0/0/1
B       172.16.1.0/24 [20/0] via 172.16.0.1, 00:00:32
        10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
B       10.1.1.0/24 [20/0] via 10.0.0.1, 00:00:31
C       10.0.0.0/30 is directly connected, Serial0/0/0 C
        192.168.0.0/24 is directly connected, Loopback0
C       192.168.1.0/24 is directly connected, Loopback1
```

ITA has routes to the loopback networks at each ISP router.

Step 5: Verify BGP on the routers.

- a. To verify the BGP operation on ITA, issue the **show ip bgp** command.

```
ITA# show ip bgp
BGP table version is 5, local router ID is 192.168.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.1.1.0/24	10.0.0.1	0		0	200 i
*> 172.16.1.0/24	172.16.0.1	0		0	300 i *>
192.168.0.0	0.0.0.0	0		32768	i
*> 192.168.1.0	0.0.0.0	0		32768	i

Step 6: Configure route filters.

- a. Check the ISP2 routing table using the **show ip route** command. ISP2 should have a route that belongs to ISP1, network 10.1.1.0.

```
ISP2# show ip route
<output omitted>
```

```

172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.0.0/30 is directly connected, Serial0/0/1
C    172.16.1.0/24 is directly connected, Loopback0
10.0.0.0/24 is subnetted, 1 subnets
B    10.1.1.0 [20/0] via 172.16.0.2, 00:05:22
B    192.168.0.0/24 [20/0] via 172.16.0.2, 00:17:45
B    192.168.1.0/24 [20/0] via 172.16.0.2, 00:17:45

```

- b. On the ITA router, configure the following access list.

```
ITA(config)# access-list 1 permit 192.168.0.0 0.0.1.255
```

- c. Apply this access list as a route filter using the **distribute-list** keyword with the BGP **neighbor** statement.

```
ITA(config)# router bgp 100
ITA(config-router)# neighbor 10.0.0.1 distribute-list 1 out
ITA(config-router)# neighbor 172.16.0.1 distribute-list 1 out
```

- d. Check the routing table for ISP2 again. The route to 10.1.1.0, ISP1, should still be in the table.

- e. Return to ITA and issue the **clear ip bgp *** command.

- f. **Step 7: Configure primary and backup routes using floating static routes.**

With bidirectional communication established with each ISP via BGP, configure the primary and backup routes. This can be done with floating static routes or BGP. a. Issue the **show ip route** command on the ITA router.

```
ITA# show ip route
<output omitted>
```

```
Gateway of last resort is not set
```

```

172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
B    172.16.0.0/30 is directly connected, Serial0/0/1
B    172.16.1.0/24 [20/0] via 172.16.0.1, 00:06:58
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
B    10.1.1.0/24 [20/0] via 10.0.0.1, 00:06:58
C    10.0.0.0/30 is directly connected, Serial0/0/0 C
192.168.0.0/24 is directly connected, Loopback0
C    192.168.1.0/24 is directly connected, Loopback1 .

```

- b. Configure static routes to reflect the policy that ISP1 is the primary provider and that ISP2 acts as the backup by specifying a lower distance metric for the route to ISP1 (210) as compared to the backup route to ISP2 (distance metric 220).

```
ITA(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.1 210
ITA(config)# ip route 0.0.0.0 0.0.0.0 172.16.0.1 220
```

- c. Verify that a default route is defined using the **show ip route** command.

```
ITA# show ip route
<output omitted>
```

```
Gateway of last resort is 10.0.0.1 to network 0.0.0.0
```

```

172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.0.0/30 is directly connected, Serial0/0/1
B    172.16.1.0/24 [20/0] via 172.16.0.1, 00:11:41
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
B    10.1.1.0/24 [20/0] via 10.0.0.1, 00:11:41

```

```

C      10.0.0.0/30 is directly connected, Serial0/0/0 C
      192.168.0.0/24 is directly connected, Loopback0
C      192.168.1.0/24 is directly connected, Loopback1 S*
0.0.0.0/0 [210/0] via 10.0.0.1

```

- d. Test this default route by creating an unadvertised loopback on the router for ISP1.

```

ISP1# config t
ISP1(config)# interface loopback 100
ISP1(config-if)# ip address 192.168.100.1 255.255.255.0

```

Step 8: Configure primary and backup routes using a default network and a static route.

Another method for configuring primary and backup routes is to use the **ip default-network** command instead of a 0.0.0.0/0 route.

- a. Remove the floating static routes configured in Step 7.

```

ITA(config)# no ip route 0.0.0.0 0.0.0.0 10.0.0.1 210
ITA(config)# no ip route 0.0.0.0 0.0.0.0 172.16.0.1 220

```

- b. The network that was added in Step 7, 192.168.100.0/24, should now be advertised on the ISP1 router. You might need to wait a few moments for BGP to advertise the new network.

```

ISP1(config)# router bgp 200
ISP1(config-router)# network 192.168.100.0 ISP1(config-router)#
end

```

- c. Make sure that the classful network 192.168.100.0 /24 appears in the ITA routing table.

```

ITA# show ip route
<output omitted>

```

Gateway of last resort is not set

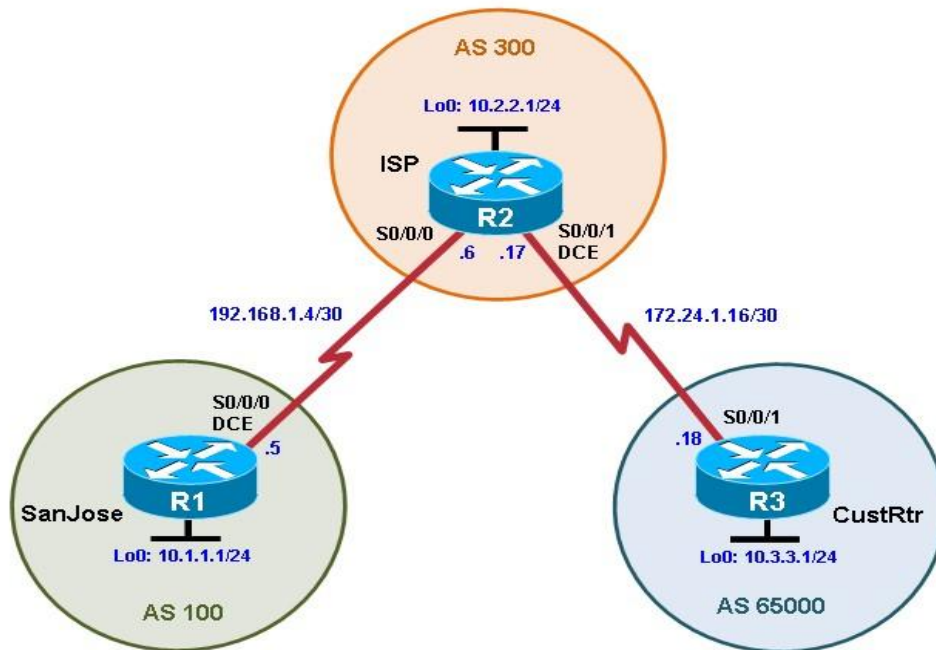
```

      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.16.0.0/30 is directly connected, Serial0/0/1
B      172.16.1.0/24 [20/0] via 172.16.0.1, 00:30:10
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks B
10.1.1.0/24 [20/0] via 10.0.0.1, 00:02:33
C      10.0.0.0/30 is directly connected, Serial0/0/0 C
      192.168.0.0/24 is directly connected, Loopback0
C      192.168.1.0/24 is directly connected, Loopback1
B      192.168.100.0/24 [20/0] via 10.0.0.1, 00:02:33

```

b) Using the AS_PATH Attribute

Topology



Objectives

- Use BGP commands to prevent private AS numbers from being advertised to the outside world.
- Use the AS_PATH attribute to filter BGP routes based on their source AS numbers.

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- Serial and console cables

Step 1: Prepare the routers for the lab.

Cable the network as shown in the topology diagram. Erase the startup configuration and reload each router to clear previous configurations.

Step 2: Configure the hostname and interface addresses.

- a. You can copy and paste the following configurations into your routers to begin.

Router R1 (hostname SanJose) `hostname SanJose`

```
! interface
Loopback0
 ip address 10.1.1.1 255.255.255.0
! interface
Serial0/0/0
 ip address 192.168.1.5 255.255.255.252
 clock rate 128000 no shutdown
```

Step 3: Configure BGP.

- a. Configure BGP for normal operation. Enter the appropriate BGP commands on each router so that they identify their BGP neighbors and advertise their loopback networks.

```
SanJose(config)# router bgp 100
SanJose(config-router)# neighbor 192.168.1.6 remote-as 300
SanJose(config-router)# network 10.1.1.0 mask 255.255.255.0
```

Verify that these routers have established the appropriate neighbor relationships by issuing the **show ip bgp neighbors** command on each router.

```
ISP# show ip bgp neighbors
BGP neighbor is 172.24.1.18, remote AS 65000, external link
  BGP version 4, remote router ID 10.3.3.1
BGP state = Established, up for 00:02:05
<output omitted>
```

```
BGP neighbor is 192.168.1.5, remote AS 100, external link
  BGP version 4, remote router ID 10.1.1.1
  BGP state = Established, up for 00:04:19
<output omitted>
```

Step 4: Remove the private AS.

- a. Display the SanJose routing table using the **show ip route** command. SanJose should have a route to both 10.2.2.0 and 10.3.3.0. Troubleshoot if necessary.

```
SanJose# show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-
2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 3 subnets
B       10.3.3.0 [20/0] via 192.168.1.6, 00:01:11 B
        10.2.2.0 [20/0] via 192.168.1.6, 00:02:16
C       10.1.1.0 is directly connected, Loopback0
        192.168.1.0/30 is subnetted, 1 subnets
C       192.168.1.4 is directly connected, Serial0/0/0
```

- b. Ping the 10.3.3.1 address from SanJose.
- c. Ping again, this time as an extended ping, sourcing from the Loopback0 interface address.

```
SanJose# ping
Protocol [ip]:
Target IP address: 10.3.3.1
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: 10.1.1.1
Type of service [0]:
```

```

Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/64/68 ms

```

- d. Check the BGP table from SanJose by using the **show ip bgp** command. Note the AS path for the 10.3.3.0 network. The AS 65000 should be listed in the path to 10.3.3.0.

```
SanJose# show ip bgp
```

```

BGP table version is 5, local router ID is 10.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i -
internal      Origin codes: i - IGP, e - EGP, ? - incomplete

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.1.1.0	0.0.0.0	0		32768	i
*> 10.2.2.0	192.168.1.6	0		0 300	i *>
10.3.3.0	192.168.1.6			0 300 65000	i

- e. Configure ISP to strip the private AS numbers from BGP routes exchanged with SanJose using the following commands.

```

ISP(config)# router bgp 300
ISP(config-router)# neighbor 192.168.1.5 remove-private-as

```

```
SanJose# ping 10.3.3.1 source lo0
```

```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.3.1, timeout is 2 seconds: Packet
sent with a source address of 10.1.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/32 ms

```

Step 5: Use the AS_PATH attribute to filter routes.

- a. Configure a special kind of access list to match BGP routes with an AS_PATH attribute that both begins and ends with the number 100. Enter the following commands on ISP.

```

ISP(config)# ip as-path access-list 1 deny ^100$ ISP(config)#
ip as-path access-list 1 permit .*

```

- b. Apply the configured access list using the **neighbor** command with the **filter-list** option.

```
ISP(config)# router bgp 300
```

Use the **clear ip bgp *** command to reset the routing information. Wait several seconds and then check the routing table for ISP. The route to 10.1.1.0 should be in the routing table.

```
ISP# show ip route
```

```
<output omitted>
```

```

172.24.0.0/30 is subnetted, 1 subnets
C    172.24.1.16 is directly connected, Serial0/0/1
10.0.0.0/24 is subnetted, 3 subnets
B    10.3.3.0 [20/0] via 172.24.1.18, 00:07:34
C    10.2.2.0 is directly connected, Loopback0

```

```
B      10.1.1.0 [20/0] via 192.168.1.5, 00:10:53
```

```
      192.168.1.0/30 is subnetted, 1 subnets
```

```
C      192.168.1.4 is directly connected, Serial0/0/0
```

- c. Check the routing table for CustRtr. It should not have a route to 10.1.1.0 in its routing table.

```
CustRtr# show ip route
```

```
<output omitted>
```

```
      172.24.0.0/30 is subnetted, 1 subnets
```

```
C      172.24.1.16 is directly connected, Serial0/0/1
```

```
      10.0.0.0/24 is subnetted, 2 subnets
```

```
C      10.3.3.0 is directly connected, Loopback0
```

```
B      10.2.2.0 [20/0] via 172.24.1.17, 00:11:57
```

- d. Return to ISP and verify that the filter is working as intended. Issue the **show ip bgp regexp ^100\$** command.

```
ISP# show ip bgp regexp ^100$
```

```
BGP table version is 4, local router ID is 10.2.2.1
```

```
Status codes: s suppressed, d damped, h history, * valid, > best, i -  
internal      Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.1.1.0	192.168.1.5	0		0 100	i

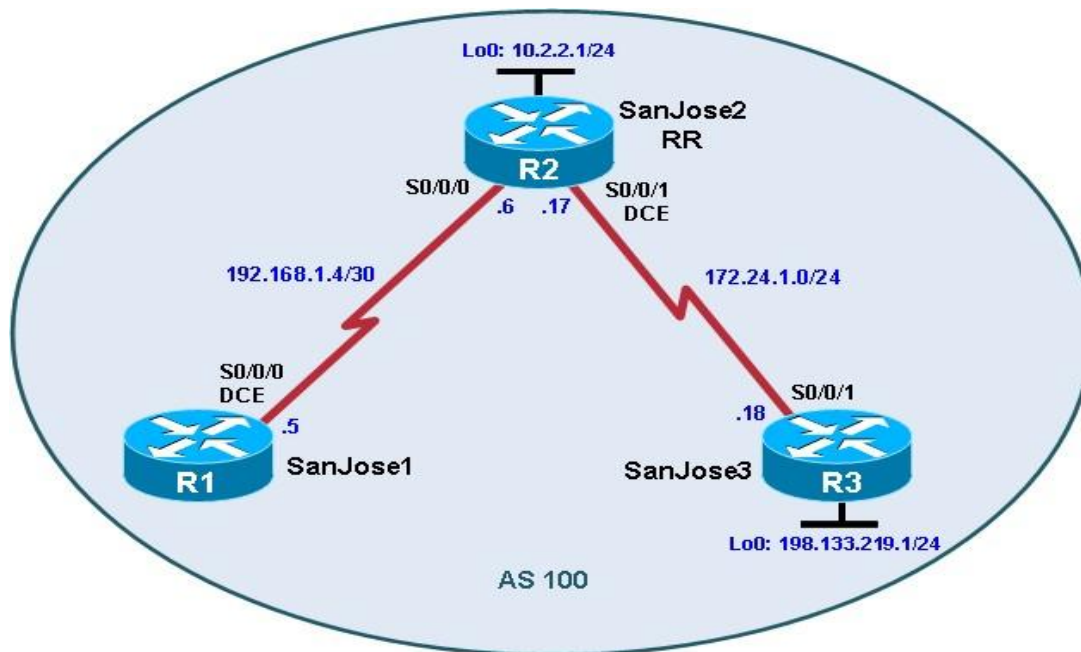
```
ip as-path access-list 1 deny ^100$
```

```
ip as-path access-list 1 permit .*
```

```
! end
```

c). BGP Route Reflectors and Route Filters

Topology



Objectives

- Configure IBGP routers to use a route reflector and a simple route filter.

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- Serial and console cables

Step 1: Prepare the routers for the lab.

Cable the network as shown in the topology diagram. Erase the startup configuration and reload each router to clear previous configurations. Do not configure Loopback 0 on SanJose3 at this time.

Step 2: Configure the hostname and interface addresses.

You can copy and paste the following configurations into your routers to begin.

Router R1 (hostname SanJose1)

```
hostname SanJose1
!
interface Serial0/0/0
 ip address 192.168.1.5 255.255.255.252
 clock rate 128000 no shutdown
```

Step 3: Configure RIPv2.

- a. Build and configure the network according to the diagram. Use RIPv2 as the IGP. Do not configure the 198.133.219.0 network under the RIP process.

```
SanJose1(config)# router rip
```

```
SanJose1(config-router)# version 2
SanJose1(config-router)# no auto-summary
SanJose1(config-router)# network 192.168.1.0
```

- b. Issue the **show ip route** command on the routers to verify that each router has a complete routing table.

```
SanJose1# show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2          E1
- OSPF external type 1, E2 - OSPF external type 2
    i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
    ia - IS-IS inter area, * - candidate default, U - per-user static
route
    o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
      172.24.0.0/24 is subnetted, 1 subnets
R      172.24.1.0 [120/1] via 192.168.1.6, 00:00:21, Serial0/0/0
      10.0.0.0/24 is subnetted, 1 subnets
R      10.2.2.0 [120/1] via 192.168.1.6, 00:00:21, Serial0/0/0
      192.168.1.0/30 is subnetted, 1 subnets
C      192.168.1.4 is directly connected, Serial0/0/0
```

Step 4: Configure IBGP peers and route reflectors.

- a. Configure the IBGP peers for BGP. Later, you will configure SanJose2 as the route reflector. However, first configure it to peer with both of the other routers.

```
SanJose2(config)# router bgp 100
SanJose2(config-router)# neighbor 192.168.1.5 remote-as 100
SanJose2(config-router)# neighbor 172.24.1.18 remote-as 100
```

Step 5: Inject a network into BGP.

- a. To observe the full effect of using a route reflector, configure SanJose3 to inject external routing information into BGP.

```
SanJose3(config)# interface loopback 0
SanJose3(config-if)# ip address 198.133.219.1 255.255.255.0
```

```
SanJose3(config-if)# router bgp 100
SanJose3(config-router)# network 198.133.219.0
```

This configuration forces SanJose3 to inject the external route 198.133.219.0 into BGP. Use the **show ip route** command to check if SanJose2 has picked up this route through BGP. SanJose2 should have a route to 198.133.219.0.

```
SanJose2# show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
    i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
    ia - IS-IS inter area, * - candidate default, U - per-user static
route
    o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```

172.24.0.0/24 is subnetted, 1 subnets
C    172.24.1.0 is directly connected, Serial0/0/1
10.0.0.0/24 is subnetted, 1 subnets
C    10.2.2.0 is directly connected, Loopback0
B    198.133.219.0/24 [200/0] via 172.24.1.18, 00:01:48
C    10.2.2.0 is directly connected, Loopback0
192.168.1.0/30 is subnetted, 1 subnets
C    192.168.1.4 is directly connected, Serial0/0/0

```

- b. Remember that SanJose1 is not configured to peer with SanJose3. To eliminate the need for a full IBGP mesh, SanJose2 must be configured as a route reflector. Issue the following commands on SanJose2:

```

SanJose2(config)# router bgp 100
SanJose2(config-router)# neighbor 192.168.1.5 route-reflector-client
SanJose2(config-router)# neighbor 172.24.1.18 route-reflector-client
*Mar  9 19:02:27.831: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Down RR client
conf ig change
*Mar  9 19:02:27.931: %BGP-5-ADJCHANGE: neighbor 172.24.1.18 Down RR client
conf ig change
*Mar  9 19:02:32.387: %BGP-5-ADJCHANGE: neighbor 172.24.1.18 Up
*Mar  9 19:02:37.507: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Up

```

- e. Verify that an IBGP cluster was successfully created by issuing the **show ip protocols** command on SanJose2. The output of this command should indicate that SanJose2 is a route reflector.

```

SanJose2# show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 26 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface          Send Recv Triggered RIP Key-chain
    Serial0/0/0         2    2
Serial0/0/1             2    2
    Loopback0           2    2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
    172.24.0.0
    192.168.1.0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 120)

```

```

Routing Protocol is "bgp 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Route Reflector for address family IPv4 Unicast, 2 clients
  Route Reflector for address family IPv6 Unicast, 2 clients
  Route Reflector for address family IPv4 MDT, 2 clients

```

Step 6: Inject a summary address into BGP.

- a. For the purpose of this lab, configure SanJose3 to inject a summary address into BGP.

```
SanJose3(config)# router bgp 100  
SanJose3(config-router)# aggregate-address 198.0.0.0 255.0.0.0
```

BGP should now send the supernet route 198.0.0.0/8 to SanJose2 with the attribute ATOMIC_AGGREGATE set.

Note: By default, BGP on Cisco routers advertises both aggregate routes and the individual component routes. If only the aggregate route is to be advertised, use the **aggregate-address network mask summary-only** command.

- b. On SanJose2, issue the following command:

```
SanJose2# show ip bgp 198.0.0.0  
BGP routing table entry for 198.0.0.0/8, version 8  
Paths: (1 available, best #1, table Default-IP-Routing-Table)  
Flag: 0x820  
  Advertised to update-groups:  
    1  
    Local, (aggregated by 100 172.24.1.18), (Received from a RR-client)  
      172.24.1.18 from 172.24.1.18 (172.24.1.18)  
      Origin IGP, metric 0, localpref 100, valid, internal, atomic-aggregate,  
best
```

- c. SanJose2 should, in turn, reflect this route to SanJose1. Check both the routing table and BGP table on SanJose1 to be sure. Both the route to 198.133.219.0 and the supernet route 198.0.0.0 should be installed in the SanJose1 routing table and the BGP table.

```
SanJose1# show ip route  
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP  
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
        ia - IS-IS inter area, * - candidate default, U - per-user static  
route  
        o - ODR, P - periodic downloaded static route
```

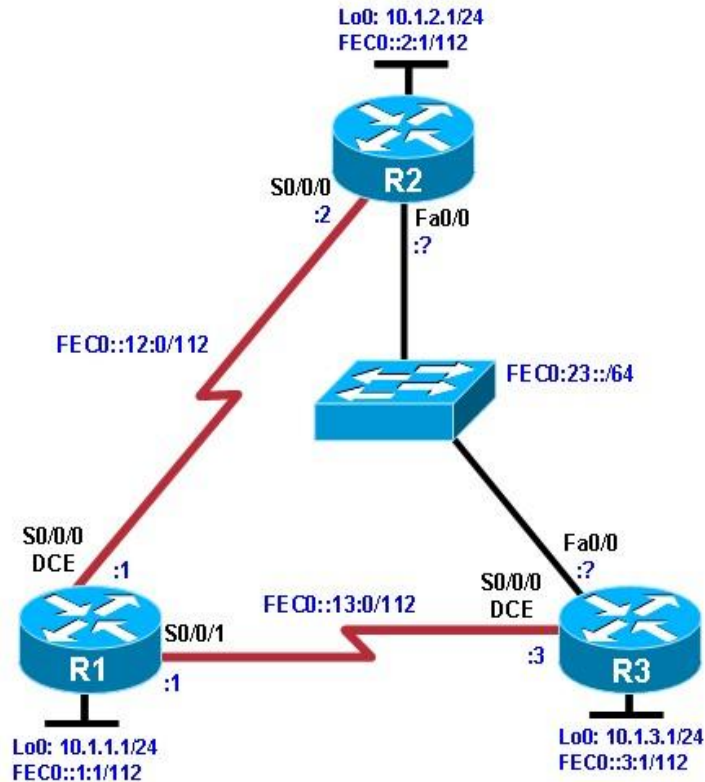
Gateway of last resort is not set

```
      172.24.0.0/24 is subnetted, 1 subnets  
R      172.24.1.0 [120/1] via 192.168.1.6, 00:00:20, Serial0/0/0  
      10.0.0.0/24 is subnetted, 1 subnets  
R      10.2.2.0 [120/1] via 192.168.1.6, 00:00:20, Serial0/0/0  
B      198.133.219.0/24 [200/0] via 172.24.1.18, 00:08:34  
      192.168.1.0/30 is subnetted, 1 subnets  
C      192.168.1.4 is directly connected, Serial0/0/0 B  
198.0.0.0/8 [200/0] via 172.24.1.18, 00:04:19
```

Practical 5

a). Configuring OSPF for IPv6

Topology



Objectives

- Configure a static IPv6 address on an interface.
- Change the default link-local address on an interface.
- Configure an EUI-64 IPv6 address on an interface.
- Enable IPv6 routing and CEF.
- Configure and verify single-area OSPFv3 operation.

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- 1 switch (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- Serial and Ethernet cables

Step 1: Prepare the routers for the lab.

Cable the network as shown in the topology diagram. Erase the startup configuration, and reload each router to clear the previous configurations.

Step 2: Configuring the hostname and loopback interfaces.

Configure the loopback interface on each router with both the IPv4 address and IPv6 address shown in the diagram.


```
Router(config)# hostname R1
R1(config)# interface loopback0
R1(config-if)# ip address 10.1.1.1 255.255.255.0 R1(config-
if)# ipv6 address FEC0::1:1/112
```

Step 3: Configure static IPv6 addresses.

- a. Configure the two serial links with IPv6 addresses. Use the **ipv6 address address/mask** command to configure the interfaces with the addresses shown in the diagram. Set the clock rates where appropriate, and bring up the interfaces.

```
R1(config)# interface serial0/0/0
R1(config-if)# ipv6 address FEC0::12:1/112
R1(config-if)# clockrate 64000 R1(config-
if)# bandwidth 64 R1(config-if)# no
shutdown
```

- b. Use **ping** to verify local subnet connectivity.

```
R1# ping FEC0::12:2
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to FEC0::12:2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/28 ms

```
R1# ping FEC0::13:3
```

Step 4: Change the link-local address on an interface.

- a. Use the **show ipv6 interface** command to look at IPv6-related properties of the router interfaces. You can also specify a type/number of an interface to see the output of that interface only.

```
R1# show ipv6 interface serial 0/0/0
Serial0/0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::219:6FF:FE23:4380
No Virtual link-local address(es):
Global unicast address(es):
FEC0::12:1, subnet is FEC0::12:0/112
Joined group address(es):
FF02::1
FF02::2
FF02::1:FF12:1
FF02::1:FF23:4380
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds
```

- b. To change the link-local address, use the **ipv6 address address link-local** command.

```
R1(config)# interface serial0/0/0
R1(config-if)# ipv6 address FE80::1 link-local
```

```
R2(config)# interface serial0/0/0
```

- d. Verify the link-local addresses with the **show ipv6 interface** command.

```
R1# show ipv6 interface serial 0/0/0
```

```

Serial0/0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::1
No Virtual link-local address(es):
Global unicast address(es):
  FEC0::12:1, subnet is FEC0::12:0/112
Joined group address(es):
  FF02::1
  FF02::2
  FF02::1:FF00:1
  FF02::1:FF12:1
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds

```

Step 5

- a. Configure IPv6 EUI-64 addresses on an interface using the **ipv6 address address/mask eui-64** command. Configure the R2 and R3 Fast Ethernet interfaces with the subnet shown in the diagram, and enable the interfaces.

```

R2(config)# interface fastEthernet 0/0
R2(config-if)# ipv6 address FEC0:23::/64 eui-64 R2(config-if)#
no shutdown

```

- b. Get the IPv6 addresses of the interfaces with the **show ipv6 interface** or **show ipv6 interface brief** command, and then ping the other side of the link.

```

R2# show ipv6 interface fastEthernet 0/0
FastEthernet0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::218:B9FF:FE92:28D8
No Virtual link-local address(es):
Global unicast address(es):
  FEC0:23::218:B9FF:FE92:28D8, subnet is FEC0:23::/64 [EUI]
Joined group address(es):
  FF02::1
  FF02::1:FF92:28D8
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds (using 17162)

```

```

FE80::218:B9FF:FE92:28D8
FEC0::2:1

```

```

R3# show ipv6 interface brief
FastEthernet0/0 [up/up]
FE80::218:B9FF:FECD:BEF0
  FEC0:23::218:B9FF:FECD:BEF0
FastEthernet0/1 [administratively down/down]
Serial0/0/0 [up/up]
  FE80::218:B9FF:FECD:BEF0

```

```

FEC0::13:3
Serial0/0/1          [administratively down/down]
Serial0/1/0          [administratively down/down]
Serial0/1/1          [administratively down/down]
Loopback0            [up/up]
FE80::218:B9FF:FECD:BEF0
FEC0::3:1

```

Step 6: Enable IPv6 routing and CEF.

use the **ipv6 unicast-routing** command in global configuration mode. To enable IPv6 CEF, use the **ipv6 cef** command. Enter these commands on all three routers.

```

R1(config)# ipv6 unicast-routing
R1(config)# ipv6 cef

```

Step7: Configure OSPFv3.

- Add all interfaces shown in the diagram into OSPF process 1, area 0. After you add the interfaces to the OSPF process, the OSPF process starts automatically. If the adjacencies do not come up after a reasonable period of time, troubleshoot using the **debug ipv6 ospf adjacency** and **debug ipv6 packet** commands. Make sure that the packets are being sent to their destination and that adjacencies are forming correctly.

```

R1(config)# interface loopback0
R1(config-if)# ipv6 ospf 1 area 0
R1(config-if)# interface serial0/0/0
R1(config-if)# ipv6 ospf 1 area 0
R1(config-if)# interface serial0/0/1
R1(config-if)# ipv6 ospf 1 area 0

```

- Verify that you have OSPFv3 neighbors with the **show ipv6 ospf neighbor** command.

```

R1# show ipv6 ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
10.1.3.1	1	FULL/ -	00:00:39	6	Serial0/0/1
10.1.2.1	1	FULL/ -	00:00:34	6	Serial0/0/0

- View the routing table on all three routers with the **show ipv6 route** command.

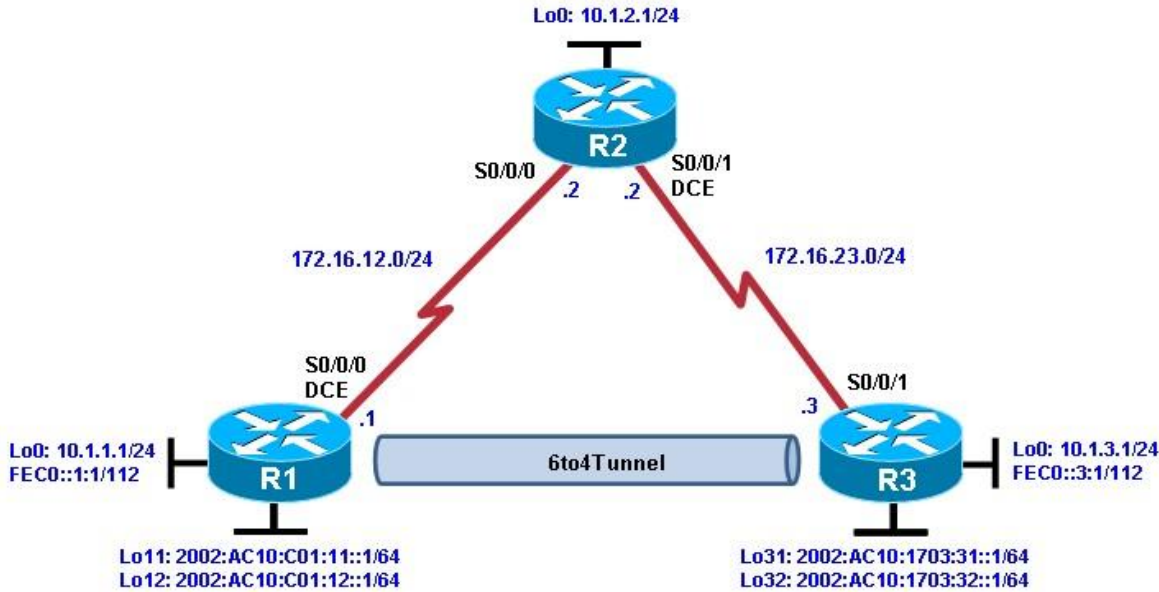
```

R1# show ipv6 route
IPv6 Routing Table - 11 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
U - Per-user Static route
    I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
    O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
    ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
    D - EIGRP, EX - EIGRP external
L   FE80::/10 [0/0]
via ::, Null0 C
FEC0::1:0/112 [0/0]
via ::, Loopback0 L
FEC0::1:1/128 [0/0]
via ::, Loopback0 O
FEC0::2:1/128
[110/1562]          via
FE80::2, Serial0/0/0 O

```

b) Configuring 6to4 Tunnels

Topology



Objectives

- Configure EIGRP for IPv4.
- Create a 6to4 tunnel.
- Configure static IPv6 routes.

Required Resources

- 3 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- Serial and console cables

Step 1: Prepare the routers for the lab.

Clear previous configurations by erasing the startup configuration and reloading the routers. After the routers are reloaded, set up the appropriate hostnames.

Step 2: Configure loopbacks and physical interfaces.

- Configure the loopback interfaces with IPv4 addresses and IPv6 addresses, where appropriate. Also configure the serial interfaces with the IPv4 addresses shown in the diagram. Set the clock rates on the appropriate interfaces and the bandwidth on all serial connections.

```
R1(config)# interface loopback0
R1(config-if)# ip address 10.1.1.1 255.255.255.0
R1(config-if)# ipv6 address FEC0::1:1/112
R1(config-if)# interface serial0/0/0
R1(config-if)# ip address 172.16.12.1 255.255.255.0
R1(config-if)# clockrate 64000 R1(config-if)#
bandwidth 64
R1(config-if)# no shutdown
```

- Verify that you have local subnet connectivity with ping.

Step 3: Configure EIGRP.

Configure EIGRP for AS 1 for the major networks 172.16.0.0 and 10.0.0.0 on all three routers. You should have full IPv4 connectivity.

```
R1(config)# router eigrp 1
R1(config-router)# no auto-summary
R1(config-router)# network 10.0.0.0
R1(config-router)# network 172.16.0.0
```

Step 4: configure a manual IPv6 6to4 tunnel.

- a. Add the following loopbacks to R1 and R3.

```
R1(config-if)# interface loopback11
R1(config-if)# ipv6 address 2002:AC10:0C01:11::1/64
R1(config-if)# interface loopback12
R1(config-if)# ipv6 address 2002:AC10:0C01:12::1/64
```

```
R3(config-if)# interface loopback31
R3(config-if)# ipv6 address 2002:AC10:1703:31::1/64
R3(config-if)# interface loopback32
R3(config-if)# ipv6 address 2002:AC10:1703:32::1/64
```

- b. Configure a 6to4 tunnel using the **interface tunnel number** command to get to the tunnel interface configuration prompt. For simplicity, use interface number 0.

```
R1(config)# interface tunnel 0
```

- c. Set the tunnel mode with the **tunnel mode ipv6ip 6to4** command. Then set up the IPv6 address with the **ipv6 address address/mask** command. The R1 address is 2002:AC10:0C01:1::1/64, because AC10:0C01 corresponds to 172.16.12.1, with 172 being AC, 16 being 10, 12 being C, and 1 being 1. The 1 after this address is just a more specific subnet, and the 1 at the end is the host address. The R3 address is 2002:AC10:1703:1::3/64. The two addresses are not in the same /64 subnet. After setting the IPv6 addresses, set the source interface for the tunnel with the **tunnel source type/number** command.

```
R1(config-if)# tunnel mode ipv6ip 6to4
R1(config-if)# ipv6 address 2002:AC10:0C01:1::1/64 R1(config-if)#
tunnel source serial0/0/0
```

- d. Now that all the tunnel settings are set, enable IPv6 routing with the **ipv6 unicast-routing** command, and set up an IPv6 static route for the whole 2002::/16 network with the global command **ipv6 route address/mask interface**, with the interface being the tunnel you just created.

```
R1(config-if)# exit
R1(config)# ipv6 unicast-routing
R1(config)# ipv6 route 2002::/16 tunnel0
```

- e. Enter the following commands on R3.

```
R3(config)# interface tunnel 0
R3(config-if)# tunnel mode ipv6ip 6to4
R3(config-if)# ipv6 address 2002:AC10:1703:1::3/64
R3(config-if)# tunnel source serial0/0/1
R3(config-if)# exit
R3(config)# ipv6 unicast-routing
R3(config)# ipv6 route 2002::/16 tunnel0
```

- f. Verify that you can ping the other side of the tunnel from R1 to R3, and ping the newly created loopback interfaces from each router .

```
R1# ping 2002:AC10:1703:1::3
```

Step 5: Configure static IPv6 routes.

- a. Static routes with a next-hop IPv6 address are created with the **ipv6 route address/mask next-hop** command. The next hop for both routers is the IPv6 address of the other end of the tunnel.

```
R1(config)# ipv6 route FEC0::3:0/112 2002:AC10:1703:1::3
```

```
R3(config)# ipv6 route FEC0::1:0/112 2002:AC10:C01:1::1
```

- b. Verify the IPv6 static routes using the **show ipv6 route** command or by pinging the remote loopback address from each router.

```
R1# show ipv6 route
```

```
IPv6 Routing Table - Default - 11 entries
```

```
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
```

```
       B - BGP, M - MIPv6, R - RIP, I1 - ISIS L1
```

```
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
```

```
EX - EIGRP external
```

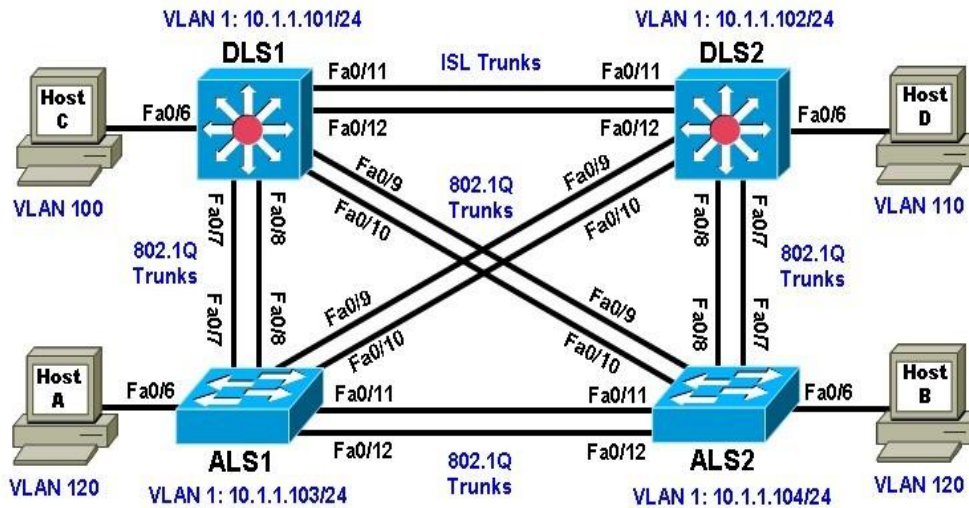
```
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
```

```
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
```

```
S    2002::/16 [1/0]
```

Practical 6,

a.) Static VLANs, VLAN Trunking, and VTP Domains and Modes Topology



Objectives

- Set up a VTP domain.
- Create and maintain VLANs.
- Configure ISL and 802.1Q trunking.

Required Resources

- 2 switches (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- 2 switches (Cisco 3560 with the Cisco IOS Release 12.2(46)SE C3560-ADVIPSERVICESK9-M image or comparable)
- 4 PCs (optional)
- Ethernet and console cables

Step 1: Prepare the switches for the lab.

Power up the switches and use the standard process for establishing a HyperTerminal console connection from a workstation to each switch in your pod. If you are connecting remotely to the switches, follow the instructions that have been supplied by your instructor.

Remove all VLAN information and configurations that may have been previously entered into the switches.

Step 2: Configure basic switch parameters.

Assign each switch a hostname and configure an IP address on the management VLAN according to the diagram. By default, VLAN 1 is used as the management VLAN.

Enter basic configuration commands on each switch according to the diagram.

Step 3: Display the switch default VLAN information.

Use the **show vlan** command in privileged mode on any switch. The following output is for a 2960 switch.

```
ALS1# show vlan
```

Step 4: Examine VTP information.

Step 5: Configure VTP on the switches.

Change the VTP domain name on DLS1 to SWLAB using the **vtp domain** command. If the VTP version defaults to 1, set it manually to version 2 using the **vtp version** command.

```
DLS1(config)# vtp domain SWLAB
Changing VTP domain name from NULL to SWLAB
```

```
DLS1(config)# vtp version 2
```

Step 6: Configure trunking.

The **show interfaces switchport** command lists the configured mode of each port in detail. The following partial sample output is for a 2960 switch on Fa0/7.

```
ALS1# show interfaces fastEthernet 0/7 switchport
```

Step 7: Verify trunk configuration. Use the show interfaces fastEthernet

0/7 switchport command on ALS2.

```
ALS2# show interfaces fastEthernet 0/7 switchport
Name: Fa0/7
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: enabled
Voice VLAN: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001 Capture Mode
Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

Step 8: Configure access ports.

A port on the 2960 switch can operate in one of three modes, and a port on the 3560 switch can operate in one of five modes. Use the **switchport mode ?** command for interface Fast Ethernet 0/6 in interface configuration mode to view the modes. The following command output is for a 2960 switch.

```
ALS1(config)# interface fastEthernet 0/6
ALS1#(config-if)# switchport mode ?
access      Set trunking mode to ACCESS unconditionally
dynamic     Set trunking mode to dynamically negotiate access or trunk mode
trunk       Set trunking mode to TRUNK unconditionally
ALS1(config)# interface fastEthernet 0/6 ALS1(config-if)#
switchport mode access
```

Step 9: Verify VTP configuration.

Before configuring the VLANs, verify the VTP configuration within the domain by using the **show vtp status** command on ALS1 and ALS2. The following sample output is from ALS1.

```
ALS1# show vtp status
```

Step 10: Configure VLANs by assigning port membership.


```
DLS1(config)# interface FastEthernet 0/6
DLS1(config-if-range)# switchport access vlan 100
Access VLAN does not exist. Creating vlan 100
VLAN 100 was created at the same time port 6 was assigned to it. Configure
```

DLS2 in the manner similar to DLS1, but this time use VLAN 110.

```
DLS2(config)# interface FastEthernet 0/6
DLS2(config-if-range)# switchport access vlan 110
% Access VLAN does not exist. Creating vlan 110
```

Issue the **show vlan** command on DLS1 to verify that VLANs 100 and 110 have been created. The output should be similar to the following output.

```
DLS1# show vlan
```

Step 11: Configure VLANs in configuration mode.

Ports still need to be assigned to VLAN 120. Port assignment to a VLAN is an interface configuration operation. Use the **switchport access vlan** command on Fast Ethernet 0/6 of ALS1 and ALS2 to configure ports for VLAN 120.

```
ALS1(config)# interface fastEthernet 0/6
ALS1(config-if)# switchport access vlan 120
ALS2(config)# interface fastEthernet 0/6
ALS2(config-if)# switchport access vlan 120
```

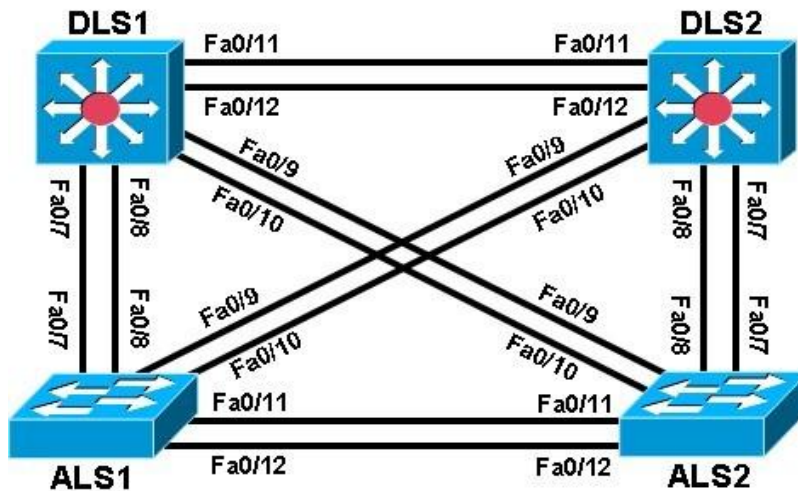
Use the **show vlan** command to verify the creation of VLAN 120, with port Fa0/6 assigned to it. The output should be similar to the following.

```
ALS1# show vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gi0/1, Gi0/2
100	VLAN0100	active	
110	VLAN0110	active	
120	VLAN0120	active	Fa0/6 1002
	fddi-default	act/unsup	1003 token-
	ring-default	act/unsup	1004 fddinet-default
	act/unsup		
1005	trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
100	enet	100100	1500	-	-	-	-	-	0	0
110	enet	100110	1500	-	-	-	-	-	0	0
120	enet	100120	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	srb	0	0

b, Configuring EtherChannel Topology



Objective

- Configure EtherChannel.

Required Resources

- 2 switches (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- 2 switches (Cisco 3560 with the Cisco IOS Release 12.2(46)SE C3560-ADVIPSERVICESK9-M image or comparable)
- Ethernet and console cables

Step 1: Prepare the switches for the lab.

Delete vlan.dat, erase the startup configuration, and reload all your switches. Refer to Lab 1-1, "Clearing a Switch," and Lab 1-2, "Clearing a Switch Connected to a Larger Network."

Step 2: Configure basic switch parameters.

- Assign each switch a hostname according to the topology diagram.
- Configure ports Fa0/7 through Fa0/12 as trunks. On the 3560 switches, you must first set the trunk encapsulation to 802.1Q. On the 2960s, only dot1q is supported, therefore the **switchport trunk encapsulation** command is unavailable, but the mode still needs to be changed to trunk.

Step 3: Configure an EtherChannel with Cisco PAgP.

- The first EtherChannel created for this lab aggregates ports Fa0/11 and Fa0/12 between ALS1 and ALS2. Make sure that you have a trunk link active for those two links with the **show interfaces trunk** command.
ALS1# **show interfaces trunk**

Port	Mode	Encapsulation	Status	Native vlan
Fa0/7	on	802.1q	trunking	1
Fa0/8	on	802.1q	trunking	1
Fa0/9	on	802.1q	trunking	1

Fa0/10	on	802.1q	trunking	1
Fa0/11	on	802.1q	trunking	1
Fa0/12	on	802.1q	trunking	1

- b. Verify that EtherChannel is working by issuing the **show etherchannel summary** command on both switches. This command displays the type of EtherChannel, the ports utilized, and port states.

ALS1# **show etherchannel summary**

```

Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended      H -
        Hot-standby (LACP only)      R - Layer3
        S - Layer2
        U - in use          f - failed to allocate aggregator
        u - unsuitable for bundling      w - waiting to be aggregated
        d - default port

```

```

Number of channel-groups in use: 1
Number of aggregators:          1

```

Group	Port-channel	Protocol	Ports
-----+-----+-----+----- 1			
Po1 (SU)	PAgP	Fa0/11 (P)	Fa0/12 (P)

- c. The **show interfaces trunk** and **show spanning-tree** commands also show the port channel as one logical link.

ALS1# **show interfaces trunk**

Port	Mode	Encapsulation	Status	Native vlan
Fa0/7	on	802.1q	trunking	1
Fa0/8	on	802.1q	trunking	1
Fa0/9	on	802.1q	trunking	1 Fa0/10
on	802.1q	trunking	1	
Po1	on	802.1q	trunking	1

<output omitted>

ALS1# **show spanning-tree**

VLAN0001

```

Spanning tree enabled protocol ieee    Root
ID    Priority    32769
      Address    0017.5a53.a380
      Cost       19
      Port       9 (FastEthernet0/9)
      Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
      Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
      Address    001d.4635.0c80
      Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time 300

```

Interface	Role	Sts	Cost	Prio.Nbr	Type

Fa0/7	Altn	BLK	19	128.7	P2p
Fa0/8	Altn	BLK	19	128.8	P2p
Fa0/9	Root	FWD	19	128.9	P2p
Fa0/10	Altn	BLK	19	128.10	P2p

Step 4: Configure an 802.3ad LACP EtherChannel.

```

ALS1(config)# interface range fastEthernet 0/7 - 8
ALS1(config-if-range)# channel-group 2 mode active
Creating a port-channel interface Port-channel 2

```

```

ALS1(config-if-range)# interface port-channel 2
ALS1(config-if)# switchport mode trunk

```

- a. Apply a similar configuration on DLS1. Verify the configuration with the **show etherchannel summary** command.

```

ALS1# show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone s - suspended      H -
        Hot-standby (LACP only)           R - Layer3
        S - Layer2
        U - in use         f - failed to allocate aggregator
        u - unsuitable for bundling        w - waiting to be aggregated
        d - default port

```

```

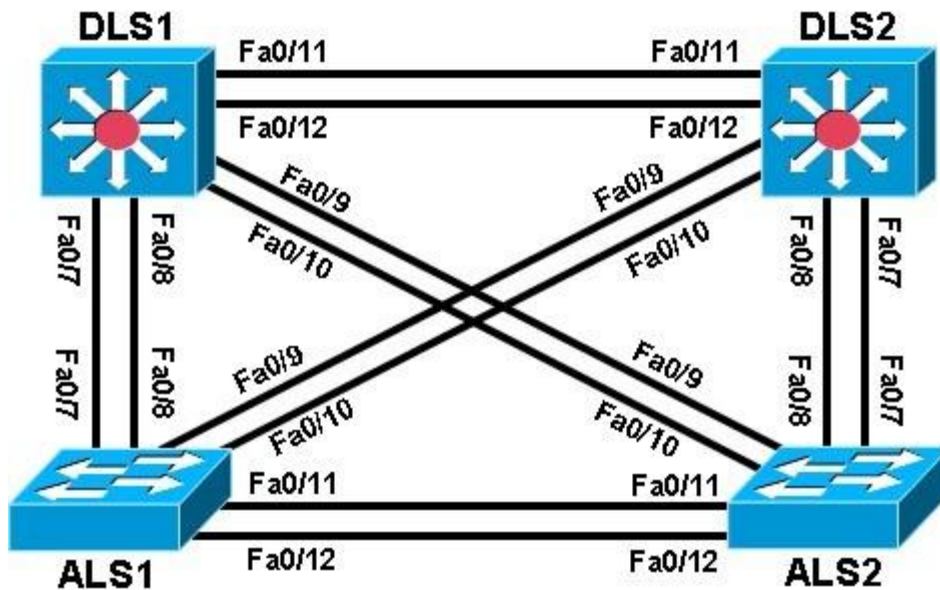
Number of channel-groups in use: 2 Number of
aggregators: 2

```

Group	Port-channel	Protocol	Ports
1	Po1 (SU)	PAgP	Fa0/11 (P) Fa0/12 (P)
2	Po2 (SU)	LACP	Fa0/7 (P) Fa0/8 (P)

Practical 7

a, Spanning Tree Protocol (STP) Default Behavior Topology



Objective

- Observe the default behavior of STP.

Required Resources

- 2 switches (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- 2 switches (Cisco 3560 with the Cisco IOS Release 12.2(46)SE C3560-ADVIPSERVICESK9-M image or comparable)

Step 1: Prepare the switches for the lab.

Refer to Lab 1-1 Clearing a Switch and Lab 1-2 Clearing a Switch Connected to a Larger Network to prepare all four switches for this lab. Cable the equipment as shown. If you are accessing your equipment remotely, ask your instructor for instructions on how to do this.

Step 2: Configure basic switch parameters.

- a. Configure the four switches as shown in the diagram with a hostname. ALS1 example:

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname ALS1
```

- b. Optionally, configure an enable secret password and console security. Configure the console line with **logging synchronous** and no timeout. ALS1 example:

```
ALS1(config)# enable secret class
ALS1(config)# line console 0
```

```

ALS1(config-line)# logging synchronous ALS1(config-
line)# exec-timeout 0 0 ALS1(configline)# password
cisco ALS1(config-line)# login

```

Step 3: Display default spanning tree information for all switches.

a. Verify IEEE 802.1D STP with the **show spanning-tree** command on DLS1.

```
DLS1# show spanning-tree
```

```

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     000a.b8a9.d680
             Cost        19
             Port        13 (FastEthernet0/11)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     000a.b8b3.d780
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time 300

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/7	Desg	FWD	19	128.9	P2p
Fa0/8	Desg	FWD	19	128.10	P2p
Fa0/9	Desg	FWD	19	128.11	P2p
Fa0/10	Desg	FWD	19	128.12	P2p
Fa0/11	Root	FWD	19	128.13	P2p
Fa0/12	Altn	BLK	19	128.14	P2p

b. Verify STP with the **show spanning-tree** command on DLS2. DLS2# **show**

```
spanning-tree
```

```

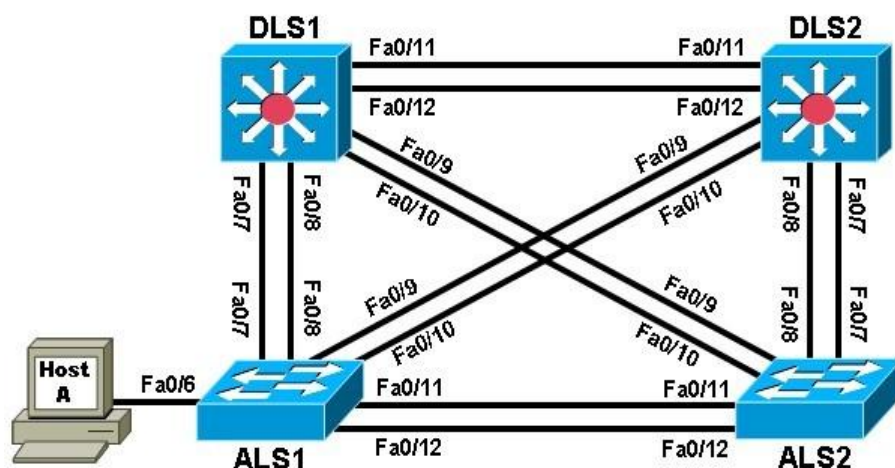
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     000a.b8a9.d680
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     000a.b8a9.d680
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time 300

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/7	Desg	FWD	19	128.9	P2p

Practical 7b, Modifying Default Spanning Tree Behavior Topology



Objective

- Observe what happens when the default spanning tree behavior is modified.

Required Resources

- 2 switches (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- 2 switches (Cisco 3560 with the Cisco IOS Release 12.2(46)SE C3560-ADVIPSERVICESK9-M image or comparable)
- 1 PC (optional) attached to switch ALS1.
- Ethernet and console cables

Step 1: Prepare the switches for the lab.

- Delete vlan.dat, erase the startup configuration, and reload all switches.
- Give each switch a hostname according to the topology diagram.

Step 2: Display default spanning tree information for all switches.

- Use the **show spanning-tree** command to check how the non-configured switches created a spanning tree. Verify which switch became the root bridge. In the topology used in this lab, DLS2 is the root bridge.

DLS1# **show spanning-tree**

```
VLAN0001
  Spanning tree enabled protocol ieee
Root ID    Priority    32769
           Address    000a.b8a9.d680
           Cost        19
           Port        13 (FastEthernet0/11)
           Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
           Address    000a.b8a9.d780
           Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time 300
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/7	Desg	FWD	19	128.9	P2p

```

Fa0/8          Desg FWD 19          128.10    P2p
Fa0/9          Desg FWD 19          128.11    P2p
Fa0/10         Desg FWD 19          128.12    P2p
Fa0/11         Root FWD 19          128.13    P2p
Fa0/12         Altn BLK 19          128.14    P2p

```

ALS1# **show spanning-tree**

VLAN0001

Spanning tree enabled protocol ieee

```

Root ID      Priority      32769
      Address      000a.b8a9.d680
      Cost          19
      Port          11 (FastEthernet0/9)
      Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec

```

```

Bridge ID    Priority      32769 (priority 32768 sys-id-ext 1)
      Address      0019.0635.5780
      Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec

```

Aging Time 300

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/7	Altn	BLK	19	128.9	P2p
Fa0/8	Altn	BLK	19	128.10	P2p
Fa0/9	Root	FWD	19	128.11	P2p
Fa0/10	Altn	BLK	19	128.12	P2p
Fa0/11	Desg	FWD	19	128.13	P2p
Fa0/12	Desg	FWD	19	128.14	P2p

- b. If you receive the following message “No spanning tree instance exists”, issue the **no shutdown** command on all interfaces.

Switch# **show spanning-tree**

No spanning tree instance exists.

Switch# **conf t**

Switch(config)# **interface range fastEthernet 0/1-24**

Switch(config-if-range)# **no shutdown**

Switch(config-if-range)# **end**

Switch# **show spanning-tree**

- c. Issue the **show interfaces trunk** command on DLS1 to verify the trunking mode, encapsulation and status for the trunk links.

DLS1# **show interfaces trunk**

Port	Mode	Encapsulation	Status	Native vlan	Fa0/7
on	802.1q	trunking			
Fa0/8	on	802.1q	trunking		8
Fa0/9	on	802.1q	trunking		8
Fa0/10	on	802.1q	trunking		8
Fa0/11	on	802.1q	trunking		2

Step 3: Configure specific switches to be primary and secondary root.

DLS2# **debug spanning-tree events**

Spanning Tree event debugging is on

b. Change DLS1 to be the primary root switch.

```
DLS1(config)# spanning-tree vlan 1 root primary
```

d. Display the running config on the new root switches, DLS1 and ALS1.

```
DLS1# show run | include span spanning-tree mode pvst
spanning-tree extend system-id spanning-tree vlan 1
priority 24576
```

```
ALS1# show run | include span
spanning-tree mode pvst spanningtree
extend system-id spanning-tree vlan 1
priority 28672
```

e. You can also view the priority modification with the **show spanning-tree** command: DLS1# **show spanning-tree**

```
VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    24577
           Address    000a.b8a9.d780
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
           Address    000a.b8a9.d780
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time 15
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/7	Desg	FWD	19	128.9	P2p
Fa0/8	Desg	FWD	19	128.10	P2p
Fa0/9	Desg	FWD	19	128.11	P2p
Fa0/10	Desg	FWD	19	128.12	P2p
Fa0/11	Desg	FWD	19	128.13	P2p
Fa0/12	Desg	FWD	19	128.14	P2p

Step 4: Change the root port using the spanning-tree port-priority command.

a. You can verify this using the **show spanning-tree** command on the non-root switch, which is DLS2. DLS2#

show spanning-tree

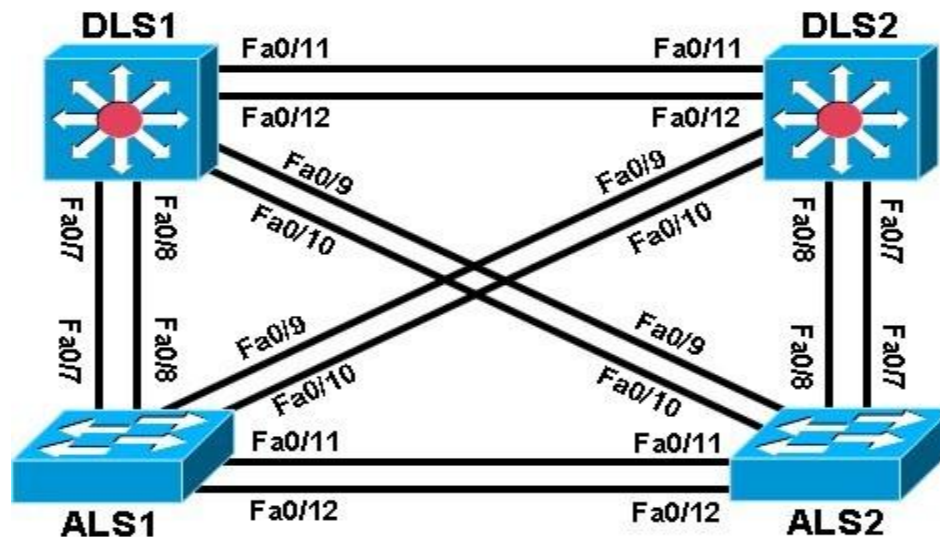
```
VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    24577
           Address    000a.b8a9.d780
           Cost        19
           Port        13 (FastEthernet0/11)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    000a.b8a9.d680
```

Practical 8

a,) Per-VLAN Spanning Tree Behavior

Topology



Objectives

- Observe the behavior of a separate spanning tree instance per VLAN.
- Change spanning tree mode to rapid spanning tree.

Required Resources

- 2 switches (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- 2 switches (Cisco 3560 with the Cisco IOS Release 12.2(46)SE C3560-ADVIPSERVICESK9-M image or comparable)
- Ethernet and console cables

Step 1: Prepare the switches for the lab.

- Delete the vlan.dat file, erase the startup configuration, and reload the switches.
- Give each switch a hostname according to the topology diagram.

DLS1 examp:

Step 2: Configure VTP.

- Configure all switches with VTP mode transparent and VTP domain CISCO. Add VLAN 10 and 20 to all of them. Use the **show vlan brief** command to view the VLAN configurations. DLS1 example:

```
DLS1(config)# vtp mode transparent
Setting device to VTP TRANSPARENT mode.
```

```
DLS1(config)# vtp domain CISCO
```

Changing VTP domain name from NULL to CISCO

```
DLS1(config)# vlan 10,20  
DLS1(config-vlan)# end
```

VLAN0001

Spanning tree enabled protocol ieee

```
Root ID      Priority      32769  
            Address      000a.b8a9.d680  
            Cost          19  
            Port          13 (FastEthernet0/11)  
            Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec  
Bridge ID    Priority      32769 (priority 32768 sys-id-ext 1)  
            Address      000a.b8a9.d780  
            Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec  
Aging Time 15
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/7	Desg	FWD	19	128.9	P2p
Fa0/8	Desg	FWD	19	128.10	P2p
Fa0/9	Desg	FWD	19	128.11	P2p
Fa0/10	Desg	FWD	19	128.12	P2p
Fa0/11	Root	FWD	19	128.13	P2p
Fa0/12	Altn	BLK	19	128.14	P2p

VLAN0010

Spanning tree enabled protocol ieee

```
Root ID      Priority      32778  
            Address      000a.b8a9.d680  
            Cost          19  
            Port          13 (FastEthernet0/11)  
            Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec  
Bridge ID    Priority      32778 (priority 32768 sys-id-ext 10)  
            Address      000a.b8a9.d780  
            Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec  
Aging Time 15
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/7	Desg	FWD	19	128.9	P2p
Fa0/8	Desg	FWD	19	128.10	P2p
Fa0/9	Desg	FWD	19	128.11	P2p
Fa0/10	Desg	FWD	19	128.12	P2p
Fa0/11	Root	FWD	19	128.13	P2p

Step 3: Assign a root switch for each VLAN.

- To change the priority for a given VLAN, use the **spanning-tree vlan number priority number** command. Configure DLS1 with priority 4096 for VLAN 10. Configure DLS2 similarly for VLAN 20.

```
DLS1(config)# spanning-tree vlan 10 priority 4096  
DLS2(config)# spanning-tree vlan 20 priority 4096
```

- b. If you look at the output of **show spanning-tree** on the four switches, you see that the port states and root switches vary on a per VLAN basis.

```
DLS1# show spanning-tree
```

```
VLAN0010
  Spanning tree enabled protocol ieee
Root ID    Priority    4106
           Address    000a.b8a9.d780
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec  Bridge ID
Priority    4106    (priority 4096 sys-id-ext 10)
           Address    000a.b8a9.d780
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec  Aging Time
30
Interface      Role Sts Cost          Prio.Nbr Type
-----
Fa0/7          Desg FWD 19          128.9   P2p
Fa0/8          Desg FWD 19          128.10  P2p
Fa0/9          Desg FWD 19          128.11  P2p
Fa0/10         Desg FWD 19          128.12  P2p
Fa0/11         Desg FWD 19          128.13  P2p
Fa0/12         Desg FWD 19          128.14  P2p
```

- a. After configuring all four switches with this command, use the **show spanning-tree** command to verify the configuration:

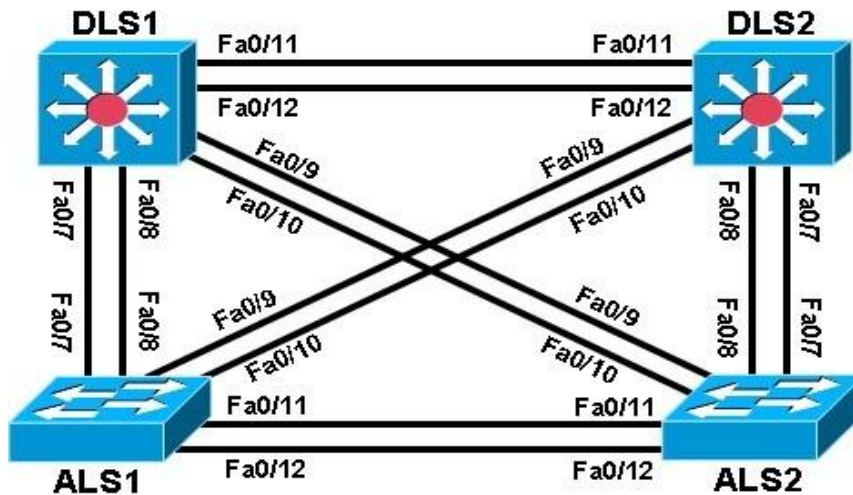
```
DLS1# show spanning-tree
```

```
Spanning tree enabled protocol rstp    Root
ID    Priority    32769
Address    000a.b8a9.d680
Cost       19
Port       13 (FastEthernet0/11)
Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
            Address    000a.b8a9.d780
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec  Aging
Time 300
```

b,) Multiple Spanning Tree

Topology



Objective

- Observe the behavior of multiple spanning tree (MST).

Required Resources

- 2 switch (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- 2 switches (Cisco 3560 with the Cisco IOS Release 12.2(46)SE C3560-ADVIPSERVICESK9-M image or comparable)
- Ethernet and console cables

Step 1: Prepare the switches for the lab.

- Delete vlan.dat file, erase the startup config, and reload the switches.
- Give each switch a hostname according to the topology diagram.

```
DLS1(config)# interface range fastEthernet 0/7 - 12
DLS1(config-if-range)# switchport trunk encapsulation dot1q
DLS1(config-if-range)# switchport mode trunk
```

Step 2: Configure VTP and VLANs.

- Configure all switches with VTP mode transparent and VTP domain CISCO. Add VLANs

10, 20, 30, 40, 50, 60, 70, 80, 90 and 100 to all of them. DLS1 example:

```
DLS1# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
DLS1(config)# vtp mode transparent
Setting device to VTP TRANSPARENT mode.
DLS1(config)# vtp domain CISCO
Changing VTP domain name from NULL to CISCO
DLS1(config)# vlan 10,20,30,40,50,60,70,80,90,100
DLS1(config-vlan)# end
```

Step 3: Display spanning tree information.

DLS1# **show spanning-tree**

VLAN0001

Spanning tree enabled protocol ieee Root ID
Priority 32769
 Address 000a.b8a9.d680
 Cost 19
 Port 13 (FastEthernet0/11)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 000a.b8a9.d780
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	-----
Fa0/7	Desg	FWD	19	128.9	P2p
Fa0/8	Desg	FWD	19	128.10	P2p
Fa0/9	Desg	FWD	19	128.11	P2p
Fa0/10	Desg	FWD	19	128.12	P2p
Fa0/11	Root	FWD	19	128.13	P2p
Fa0/12	Altn	BLK	19	128.14	P2p

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	-----
Fa0/7	Desg	FWD	19	128.9	P2p
Fa0/8	Desg	FWD	19	128.10	P2p
Fa0/9	Desg	FWD	19	128.11	P2p
Fa0/10	Desg	FWD	19	128.12	P2p
Fa0/11	Root	FWD	19	128.13	P2p
Fa0/12	Altn	BLK	19	128.14	P2p

VLAN0020

Spanning tree enabled protocol ieee Root ID
Priority 32788
 Address 000a.b8a9.d680
 Cost 19
 Port 13 (FastEthernet0/11)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

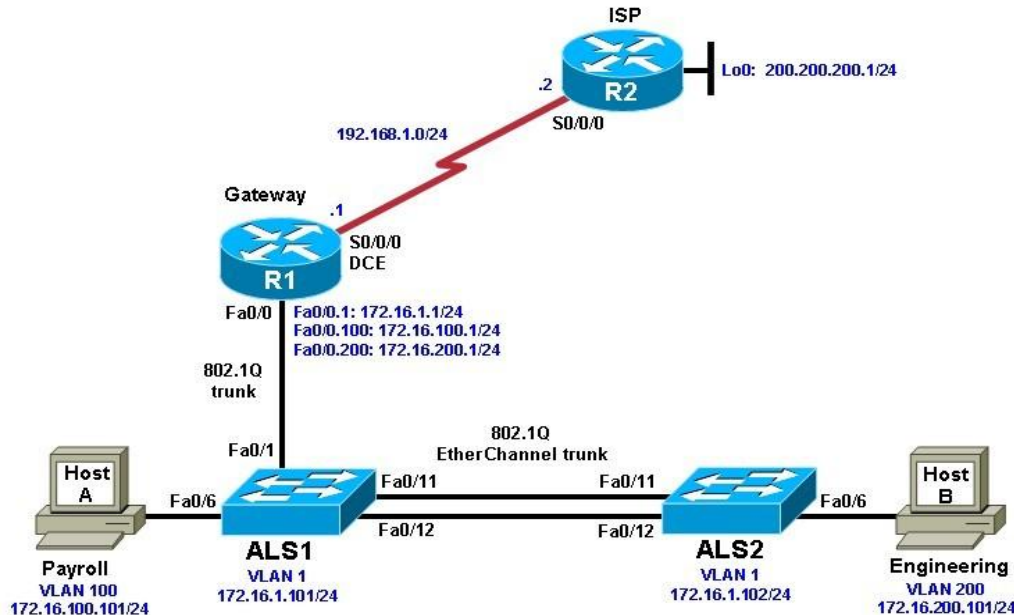
Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)
 Address 000a.b8a9.d780
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	-----
Fa0/7	Desg	FWD	19	128.9	P2p
Fa0/8	Desg	FWD	19	128.10	P2p
Fa0/9	Desg	FWD	19	128.11	P2p
Fa0/10	Desg	FWD	19	128.12	P2p
Fa0/11	Root	FWD	19	128.13	P2p

Practical 9

a, Inter-VLAN Routing with an External Router

Topology



Objective

- Configure inter-VLAN routing using an external router, also known as a router on a stick.

Required Resources

- 2 routers (Cisco 1841 with Cisco IOS Release 12.4(24)T1 Advanced IP Services or comparable)
- 2 switches (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- Serial and Ethernet cables

Step 1: Prepare the switches and routers for the lab.

- a. Cable the network as shown in the topology diagram. On each switch, erase the startup configuration, delete the vlan.dat file, and reload the switches. Refer to Lab 1-1, "Clearing a Switch" and Lab 1-2, "Clearing a Switch Connected to a Larger Network" to prepare the switches for this lab.
- b. Erase the startup configuration and reload the routers.

Step 3: Configure the routers.

- a. Configure the ISP router for communication with your gateway router. The static route used for the internal networks provides a path for the local network from the ISP. In addition, configure a loopback interface on the ISP router to simulate an external network.

```
Router(config)# hostname ISP
ISP(config)# interface Loopback0
ISP(config-if)# ip address 200.200.200.1 255.255.255.0
ISP(config-if)# interface Serial0/0/0
```

```
ISP(config-if)# ip address 192.168.1.2 255.255.255.0
ISP(config-if)# no shutdown
ISP(config-if)# exit
ISP(config)# ip route 172.16.0.0 255.255.0.0 192.168.1.1
```

Step 4: Configure the switches.

- a. Configure the switch hostnames and IP addresses on the management VLAN according to the diagram. By default, VLAN 1 is used as the management VLAN. Create a default gateway on both access layer switches using the **ip default-gateway ip_address** command. The following is a sample configuration for switch ALS1.

```
Switch(config)# hostname ALS1
ALS1(config)# interface vlan 1
ALS1(config-if)# ip address 172.16.1.101 255.255.255.0
ALS1(config-if)# no shutdown
ALS1(config-if)# exit
ALS1(config)# ip default-gateway 172.16.1.1
```

- a. Verify that the only existing VLANs are the built-in VLANs. Issue the **show vlan** command from privileged mode on both access layer switches. ALS1# **show vlan**

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gi0/1, Gi0/2
1002 fddi-default	act/unsup	1003 tokenring-
default	act/unsup	1004 fddinet-default
act/unsup 1005 trnet-default		act/unsup
VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrdgMode Trans1		
Trans2		
1 enet 100001 1500 - - - - - 0 0		
1002 fddi 101002 1500 - - - - - 0		
0		

Step 6: Configure trunk links and EtherChannel on switches.

- a. Use the Fast Ethernet 0/11 and 0/12 ports of ALS1 and ALS2 to create an EtherChannel trunk between the switches.

```
ALS1# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ALS1(config)# interface range fastEthernet 0/11 - 12
ALS1(config-if-range)# switchport mode trunk
ALS1(config-if-range)# channel-group 1 mode desirable
ALS1(config-if-range)# end
```

Step 7: Configure VTP.

- a. Use the **show vtp status** command to verify the ALS1 VTP configuration and that client ALS2 has learned the new VTP domain information from ALS1.

```
ALS1# show vtp status
VTP Version                : running VTP2
Configuration Revision      : 1
Maximum VLANs supported locally : 255
Number of existing VLANs    : 5
VTP Operating Mode         : Server VTP
Domain Name                 : SWLAB
VTP Pruning Mode           : Disabled
VTP V2 Mode                : Enabled
VTP Traps Generation       : Disabled
MD5 digest                  : 0x6A 0x1A 0x90 0xA3 0x10 0xCE 0x86 0xFA
Configuration last modified by 172.16.1.101 at 2-28-10 00:36:24 Local
updater ID is 172.16.1.101 on interface V11 (lowest numbered VLAN
interface found)
```

```
ALS2# show vtp status
VTP Version                : running VTP2
Configuration Revision      : 1
Maximum VLANs supported locally : 255
Number of existing VLANs    : 5
VTP Operating Mode         : Client
VTP Domain Name            : SWLAB
VTP Pruning Mode           : Disabled
VTP V2 Mode                : Enabled
VTP Traps Generation       : Disabled
MD5 digest                  : 0x6A 0x1A 0x90 0xA3 0x10 0xCE 0x86 0xFA
```

Step 8: Configure

VLANs and switch access ports.

- a. Configure the VLAN 100 named Payroll and VLAN 200 named Engineering on VTP server ALS1.

```
ALS1(config)# vlan 100
ALS1(config-vlan)# name Payroll
ALS1(config-vlan)#
ALS1(config-vlan)# vlan 200
ALS1(config-vlan)# name Engineering
```

- b. Use the **show vlan brief** command on ALS2 to verify that ALS2 has learned the new VLANs from ALS1.

```
ALS2# show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/6, Fa0/7, Fa0/8		Fa0/9, Fa0/10, Fa0/13, Fa0/14
		Fa0/15, Fa0/16, Fa0/17,
Fa0/18		Fa0/19, Fa0/20, Fa0/21,
Fa0/22		Fa0/23, Fa0/24, Gi0/1, Gi0/2
100 Payroll	active	
200 Engineering	active	
1002 fddi-default	act/unsup	

```

1003 trcrf-default          act/unsup 1004 fddinetdefault
    act/unsup 1005 trbrf-default          act/unsup

```

%Portfast has been configured on FastEthernet0/6 but will only have effect when the interface is in a non-trunking mode.

```

Fa0/19                      Fa0/16, Fa0/17, Fa0/18,
                             Fa0/20, Fa0/21, Fa0/22,
Fa0/23                      Fa0/24, Gi0/1, Gi0/2

```

```

100 Payroll                active Fa0/6
200 Engineering            active
1002 fddi-default          act/unsup
1003 trcrf-default         act/unsup 1004 fddinetdefault
    act/unsup 1005 trbrf-default          act/unsup

```

```
1002 act/unsup
```

9: Configure ALS1 trunking to the Gateway router.

Step 10: Configure the Gateway router Fast Ethernet interface for VLAN trunking.

- b. Use the **show ip interface brief** command to verify the interface configuration and status.

```
Gateway# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status
FastEthernet0/0	unassigned	YES	unset	up
FastEthernet0/1.1	172.16.1.1	YES	manual	up
FastEthernet0/1.100	172.16.100.1	YES	manual	up
FastEthernet0/1.200	172.16.200.1	YES	manual	up
FastEthernet0/1	unassigned	YES	unset	administratively down
Serial0/0/0	192.168.1.1	YES	manual	up
Serial0/0/1	unassigned	YES	unset	administratively down

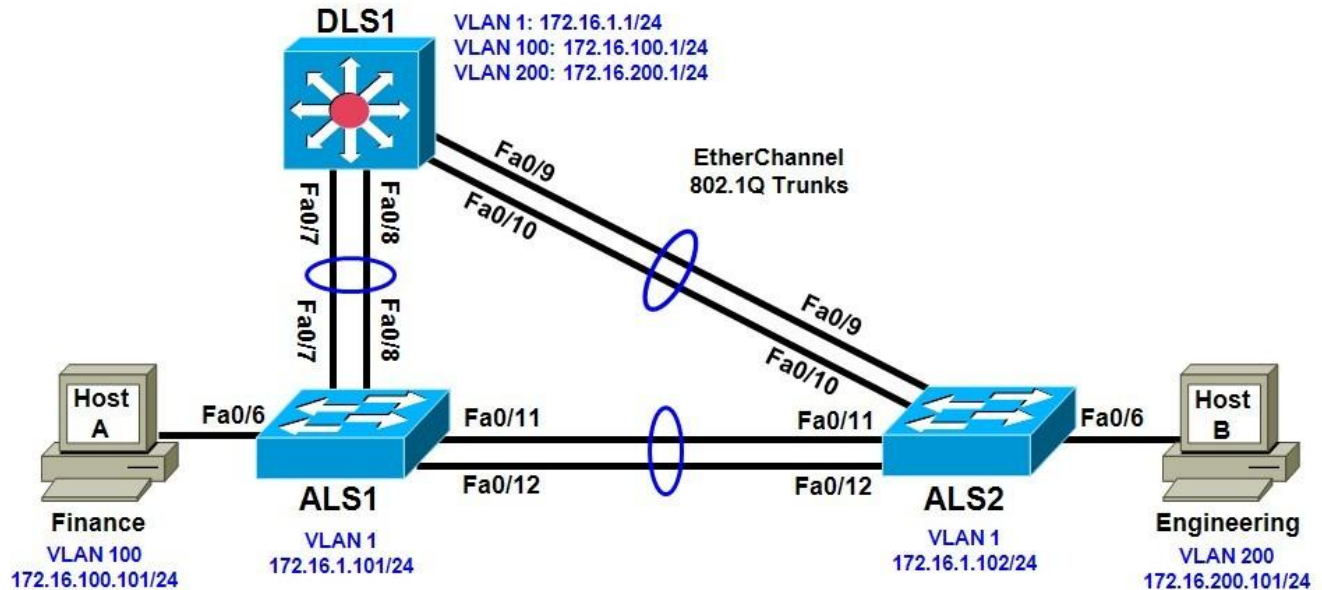
- c. Use the **show interfaces description** command to verify the interface status and description assigned.

```
Gateway# show interfaces description
```

Interface	Status	Protocol	Description
Fa0/0	up	up	
Fa0/0.1	up	up	Management VLAN 1
Fa0/0.100	up	up	Payroll VLAN 100
Fa0/0.200	up	up	Engineering VLAN 200
Fa0/1	admin down	down	
Se0/0/0	up	up	
Se0/0/1	admin down	down	

b, Inter-VLAN Routing with an Internal Route Processor and Monitoring CEF Functions

Topology



Objective

- Route between VLANs using a 3560 switch with an internal route processor using Cisco Express Forwarding (CEF).

Required Resources

- 2 switches (Cisco 2960 with the Cisco IOS Release 12.2(46)SE C2960-LANBASEK9-M image or comparable)
- 1 switch (Cisco 3560 with the Cisco IOS Release 12.2(46)SE C3560-ADVIPSERVICESK9-mz image or comparable)
- Ethernet and console cables

Step 1: Prepare the switches for the lab.

Erase the startup configuration, delete the vlan.dat file, and reload the switches.

Step 2: Configure basic switch parameters.

- Configure the hostname, password, and optionally, Telnet access on each switch.

```
Switch(config)# hostname ALS1
ALS1(config)# enable secret cisco
ALS1(config)# line vty 0 15
ALS1(config-line)# password cisco
ALS1(config-line)# login
Switch(config)# hostname ALS2
ALS2(config)# enable secret cisco
ALS2(config)# line vty 0 15
```

```
ALS2(config-line)# password cisco
ALS2(config-line)# login
```

```
Switch(config)# hostname DLS1
DLS1(config)# enable secret cisco
DLS1(config)# line vty 0 15
DLS1(config-line)#password cisco DLS1(config-line)#
login
```

- b. Configure management IP addresses on VLAN 1 for all three switches according to the diagram.

```
ALS1(config)# interface vlan 1
ALS1(config-if)# ip address 172.16.1.101 255.255.255.0
ALS1(config-if)# no shutdown
```

```
ALS2(config)# interface vlan 1
ALS2(config-if)# ip address 172.16.1.102 255.255.255.0
ALS2(config-if)# no shutdown
```

```
DLS1(config)# interface vlan 1
DLS1(config-if)# ip address 172.16.1.1 255.255.255.0
DLS1(config-if)# no shutdown
```

- c. Configure default gateways on the access layer switches. The distribution layer switch will not use a default gateway, because it acts as a Layer 3 device. The access layer switches act as Layer 2 devices and need a default gateway to send management VLAN traffic off of the local subnet.

```
ALS1(config)# ip default-gateway 172.16.1.1
```

```
ALS2(config)# ip default-gateway 172.16.1.1
```

Step 3: Configure trunks and EtherChannels between switches.

To distribute VLAN and VTP information between the switches, trunks are needed between the three switches. Configure these trunks according to the diagram. EtherChannel is used for these trunks.

EtherChannel allows you to utilize both Fast Ethernet interfaces that are available between each device, thereby doubling the bandwidth.

- a. Configure the trunks and EtherChannel from DLS1 to ALS1. The **switchport trunk encapsulation [isl | dot1q]** command is used because this switch also supports ISL encapsulation.

```
DLS1(config)# interface range fastEthernet 0/7 - 8
DLS1(config-if-range)# switchport trunk encapsulation dot1q
DLS1(config-if-range)# switchport mode trunk
DLS1(config-if-range)# channel-group 1 mode desirable
```

Creating a port-channel interface Port-channel 1

- b. Configure the trunks and EtherChannel from DLS1 to ALS2. DLS1(config)# **interface range fastEthernet 0/9 - 10**

```
DLS1(config-if-range)# switchport trunk encapsulation dot1q
DLS1(config-if-range)# switchport mode trunk
DLS1(config-if-range)# channel-group 2 mode desirable
```

Creating a port-channel interface Port-channel 2

- c. Configure the trunks and EtherChannel between ALS1 and DLS1, and for the trunks and EtherChannel between ALS1 and ALS2.

```

ALS1(config)# interface range fastEthernet 0/11 - 12
ALS1(config-if-range)# switchport mode trunk
ALS1(config-if-range)# channel-group 1 mode desirable

```

Creating a port-channel interface Port-channel 1

```

ALS1(config-if-range)# exit
ALS1(config)# interface range fastEthernet 0/7 - 8
ALS1(config-if-range)# switchport mode trunk
ALS1(config-if-range)# channel-group 2 mode desirable

```

Creating a port-channel interface Port-channel 2

- d. Configure the trunks and EtherChannel between ALS2 and DLS1, and for the trunks and EtherChannel between ALS2 and ALS1.

```

ALS2(config)# interface range fastEthernet 0/11 - 12
ALS2(config-if-range)# switchport mode trunk
ALS2(config-if-range)# channel-group 1 mode desirable

```

Creating a port-channel interface Port-channel 1

```

ALS2(config-if-range)# exit
ALS2(config)# interface range fastEthernet 0/9 - 10
ALS2(config-if-range)# switchport mode trunk
ALS2(config-if-range)# channel-group 2 mode desirable

```

Creating a port-channel interface Port-channel 2

- e. Verify trunking between DLS1, ALS1, and ALS2 using the **show interface trunk** command on all switches.

```

DLS1# show interface trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Po1	on	802.1q	trunking	1
Po2	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Po1	1-4094
Po2	1-4094

Port	Vlans allowed and active in management domain
Po1	1
Po2	1

Port	Vlans in spanning tree forwarding state and not pruned
Po1	1
Po2	1

Step 4: Configure VTP on ALS1 and ALS2.

- a. Change the VTP mode of ALS1 and ALS2 to client.

```

ALS1(config)# vtp mode client

```